

DisCo-CFL: Calibrated, Fair and Explainable Clustered Federated Learning via Disattenuated Class-Conditional Gradient Signatures

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Abstract

Clustered federated learning (CFL) trains one model per group of clients that share a data distribution. A recent survey of the field [Ben Ali et al., 2026] identifies several open problems: server-side similarities are computed on high-dimensional, round-dependent model updates; clients tend to be grouped by label distribution rather than by concept; the number of clusters is not determined in a principled way; quantity skew breaks existing methods; and the fairness, accountability and transparency of the clustering decision are almost never studied. We propose *DisCo-CFL* (Disattenuated Class-conditional CFL). At a frozen reference model, every client uploads once a count-sketches mean gradient for each class it holds, computed on two random halves of that class. From the two halves the server estimates the reliability of every class signature and applies Spearman’s correction for attenuation, which yields a *calibrated* similarity: it is invariant to label skew and quantity skew by construction, and consistent for the population cosine between class-conditional gradients even under differential-privacy noise. Clusters are formed with a class-wise bottleneck linkage and a single threshold on this calibrated scale, so the number of clusters is found automatically. We prove (i) exact invariance to target and quantity skew, (ii) a finite-sample concentration bound for the calibrated similarity, (iii) safety (conflicting clients are never merged), maximality and exact recovery with a sample complexity that does not depend on cluster size, which is a formal minority-fairness guarantee, (iv) a minimax lower bound that matches the upper bound up to a factor $1/\Delta$ and logarithmic terms, (v) an (ϵ, δ) differential-privacy guarantee with an explicit privacy-utility cost, and (vi) fidelity of the class-level explanations produced by the server. Beyond accuracy, DisCo-CFL outputs per-client assignment certificates with confidence intervals, an audit log of every merge decision, flags for low-evidence and anomalous clients, and an automatic diagnosis of the type of heterogeneity. The deterministic core of all proofs (25 theorems) is machine-checked in Lean 4. We evaluate on Fashion-MNIST, MNIST, SVHN and CIFAR-10 (618 training runs) under concept shift on features and targets, label skew, quantity skew, minority groups, varying shift strength, 2–8 groups, partial participation, 200 clients, newcomers and privacy noise. We compare with nine baselines: FedAvg, local training, FedAvg with fine-tuning, IFCA, FeSEM, CFL, FL+HC, PACFL, and an oracle. Without knowing K , DisCo-CFL recovers the ground-truth groups (mean ARI 0.94–1.00) in the rotation, label-swap, mixed and minority scenarios on Fashion-MNIST, MNIST and SVHN. It merges on average 0.1% of the conflicting client pairs there, against 7.7% (FeSEM) to 84% (FL+HC) for baselines given the true K . Its accuracy is within 0.1 points of training on the true groups, and paired Wilcoxon tests show it significantly better than every baseline on mean, concept and worst-group accuracy ($p < 0.01$). On CIFAR-10 the reference model needs a longer warm-up, and a baseline that fragments the federation into 35 clusters keeps a higher local accuracy.

1 Introduction

Federated learning (FL) trains a model across clients that keep their data local [McMahan et al., 2017, Kairouz et al., 2021]. When client data are not identically distributed, a single global model can be a poor fit for everyone [Zhao et al., 2018]. *Clustered* FL relaxes the single-model objective: clients are partitioned into groups with homogeneous distributions and each group trains its own model [Sattler et al., 2021, Ghosh et al., 2020, Briggs et al., 2020]. A recent survey [Ben Ali et al., 2026] organises the field into server-side, client-side and metadata-based methods. It shows that each family trades privacy, computation, communication and the stability of the clustering metric differently, and it lists open challenges. Our work is driven by five of them.

1. **What to cluster on.** Server-side methods compare full, high-dimensional model updates whose geometry changes from round to round. Several studies report that CFL methods tend to group clients by *label* similarity [Guo et al., 2024], which fragments the data even when clients share the same concept.
2. **Number of clusters.** Every method has a hyperparameter that controls K directly or indirectly, and “fully automatic and stable cluster determination in CFL remains an open research challenge” [Ben Ali et al., 2026].
3. **Quantity skew** is almost never evaluated, even though existing CFL algorithms are known to struggle with it [Ben Ali et al., 2025].
4. **Privacy** is rarely built into the clustering step. Metadata-based methods improve efficiency by sharing statistics that carry a moderate to high privacy risk.
5. **Evaluation is performance-only.** Methods are compared on accuracy. The *quality* of the clustering decision is rarely scrutinised, and neither is its fairness towards small groups or its explainability to the participants.

Our idea. At a fixed reference model w , the expected gradient of the loss restricted to class c depends on a client only through its *class-conditional* distribution $\mathcal{P}_i(X | Y = c)$. It is therefore unaffected by the client’s label proportions and dataset size. Both kinds of concept shift change it: a feature shift (e.g. rotated inputs) changes $\mathcal{P}(X | Y = c)$ for every class, and a target shift (e.g. swapped labels) changes it for the affected classes. Estimated class-conditional gradients, however, are noisy, and noise *attenuates* cosine similarities: two clients with the same concept but few samples look dissimilar. We borrow a century-old idea from psychometrics, Spearman’s correction for attenuation [Spearman, 1904, 1910, Brown, 1910]. Every client splits the samples of each class into two halves, and the agreement between the halves measures the reliability of the signature. Dividing the cross-client cosine by the geometric mean of the two reliabilities gives an estimator of the *population* similarity, on a scale where 1 means “same concept” for every client regardless of how much data it has or how much privacy noise was added.

Contributions.

- **Algorithm.** DisCo-CFL is a one-shot, server-side CFL algorithm. It uses sketched class-conditional gradient signatures, split-half disattenuation, and threshold clustering with a class-wise *bottleneck* linkage and evidence-ordered merging. K is determined by one threshold on a calibrated scale (Section 4).
- **Theory** (Section 5): exact invariance; a concentration bound; safety, maximality and exact recovery with a size-independent sample complexity; a minimax lower bound; differential privacy

with an explicit utility cost; explanation fidelity; and Bayes optimality of prior-corrected cluster models. The deterministic core of every proof (25 theorems) is machine-checked in Lean 4 with Mathlib (Appendix B).

- **Properties beyond accuracy** (Section 6), which we organise along fairness, accountability, transparency and privacy: minority-group protection, per-client assignment certificates, an audit log, anomaly flags, class-level explanations and an automatic heterogeneity diagnosis.
- **Evaluation** (Section 7) on four datasets against nine baselines covering all families of the taxonomy. The scenarios combine concept shift on features and targets, label skew, quantity skew, imbalanced groups, shift strength, number of groups, partial participation, scale, newcomers and DP noise. Beyond accuracy we report clustering quality, fairness, explanation fidelity and certificate calibration, with significance tests.

2 Related work and positioning

We follow the taxonomy of the recent survey by Ben Ali et al. [2026]. It separates *core* CFL (clustering to handle statistical heterogeneity by training several models) from operational uses of clustering, and within core CFL it distinguishes server-side, client-side and metadata-based methods. DisCo-CFL is core CFL. We review the three families, then the neighbouring lines of work from which DisCo-CFL borrows or with which it can be confused, and close with an explicit positioning (Table 1).

2.1 Server-side CFL

Server-side methods cluster what clients already send in FedAvg, namely model weights or updates. Ghosh et al. [2019] apply trimmed k -means to locally trained models. CFL, often called MTCFL [Sattler et al., 2021], recursively bipartitions a cluster by the cosine similarity of the clients’ updates once its joint objective is near a stationary point; a theorem guarantees that a correct bipartition exists when several distributions are present. FL+HC [Briggs et al., 2020] applies one-shot hierarchical clustering after a FedAvg warm-up and was the first to build benchmarks from a non-IID taxonomy. FlexCFL [Duan et al., 2021] and ICFL [Yan et al., 2023] reduce the cost of comparing high-dimensional updates with truncated SVD. FLACC [Mehta and Shao, 2023] merges clusters greedily, FPFC [Yu et al., 2024] penalises pairwise model distances, and CFL-GT [Liu et al., 2024] uses gradient trajectories. StoCFL [Zeng et al., 2025] compares gradients of a frozen model under dynamic participation. FeSEM [Long et al., 2023] runs an EM-like assignment of clients to the nearest of K centres in parameter space. The common signal of these methods is a whole-model update, whose geometry changes across rounds, mixes the effect of label proportions with that of concepts, and depends on the client’s sample size. The survey lists the instability of these metrics and the choice of K as open problems [Ben Ali et al., 2026, Sec. 4.3.1, 4.3.4].

2.2 Client-side CFL

In client-side CFL the server broadcasts K models and each client selects the one with the lowest training loss: HypCluster [Mansour et al., 2020] and IFCA [Ghosh et al., 2020], the latter with a convergence analysis. FLSC [Li et al., 2021a] lets clients join several clusters, FedSoft [Ruan and Joe-Wong, 2022] and FedEM [Marfoq et al., 2021] learn mixture weights, and FedRC [Guo et al., 2024] and HCFL+ [Guo et al., 2025] combine soft assignment with robust bi-level objectives and split/merge rules. FedDrift [Jothimurugesan et al., 2023] handles concept drift over time,

FedCAM [Ma et al., 2023] adds cluster-specific components to a global model, and AutoCFL [Gong et al., 2024] adapts local training to imbalanced data. Client-side methods need a fixed or initial K , download K models per round, and are sensitive to initialisation [Ben Ali et al., 2026]. Loss-based assignment also reacts to label proportions: a client whose labels are concentrated on a few classes prefers the model trained on a similar label mix.

2.3 Metadata-based CFL

Metadata-based methods share additional statistics. k-FED [Dennis et al., 2021] sends local k -means centroids of the data. FLT [Jamali-Rad et al., 2022] sends autoencoder embeddings, PACFL [Vahidian et al., 2023] principal subspaces of the raw data, FLIS [Morafah et al., 2023] predictions on a public dataset, MD-ICFL [Zhang et al., 2024] label-weighted model outputs, pp-CFL [Luo et al., 2024] noisy label distributions, and TDCFL [Ren et al., 2025] and SnapCFL [Cheng et al., 2025] other data summaries. These signals are stable across rounds and cheap to cluster, but the survey rates most of them as a moderate to high privacy risk [Ben Ali et al., 2026, Table 4]. Only pp-CFL adds a formal privacy mechanism.

2.4 Class-conditional comparisons and quantity skew

Two recent works also compare clients class by class. FedDAG [Pramanik et al., 2026] fuses gradient similarity with principal angles between class-wise *data* subspaces, weights classes by their frequency, and selects the number of clusters with a compactness criterion. FedCCFA [Chen et al., 2024] clusters class-specific classifier vectors with DBSCAN under concept drift. For quantity skew, CORNFLQS [Ben Ali et al., 2025] alternates weight-based and loss-based clustering. DisCo-CFL shares the intuition that concepts are best compared class by class. It differs in what is compared: expected class-conditional gradients at a frozen model, which are functionals of $\mathcal{P}(X | Y)$ alone (Proposition 1), rather than data subspaces or classifier rows. It also differs in how noise is handled: the similarity is disattenuated, so its population value for clients that share a concept is exactly 1 whatever the sample size or privacy noise. The number of clusters then follows from one threshold with a fixed meaning.

2.5 Neighbouring lines of work

Heterogeneity-robust and personalised FL. FedProx [Li et al., 2020a] and SCAF-FOLD [Karimireddy et al., 2020] keep a single model and correct client drift. Personalised FL [Tan et al., 2023] adapts one model per client, through meta-learning (Per-FedAvg, Fallah et al., 2020), regularisation (pFedMe, T. Dinh et al., 2020; Ditto, Li et al., 2021b), shared representations (FedRep, Collins et al., 2021) or class prototypes (FedProto, Tan et al., 2022). CFL assumes a few distinct distributions instead of one distribution with local deviations [Ben Ali et al., 2026]. We include FedAvg with local fine-tuning as the representative personalisation baseline.

Label-distribution skew. Logit calibration counters label skew in a single model (FedLC, Zhang et al., 2022), in the spirit of label-shift correction [Lipton et al., 2018] and logit adjustment [Menon et al., 2021]. DisCo-CFL uses the same Bayes argument at inference, but inside a concept cluster (Proposition 4), and uses it to justify why label skew should not create clusters.

Gradient similarity for grouping tasks. Multi-task learning groups tasks by gradient affinity [Fifty et al., 2021] and resolves gradient conflicts [Yu et al., 2020]; the gradient signal-to-noise

ratio has been linked to generalisation [Liu et al., 2020]. None of these corrects similarities for estimation noise.

Measurement error. Spearman’s correction for attenuation [Spearman, 1904, 1910, Brown, 1910] is standard in psychometrics and meta-analysis [Muchinsky, 1996]. To our knowledge it has not been used to calibrate client similarities in FL.

Privacy and communication. DP-FedAvg [McMahan et al., 2018] and DP-SGD [Abadi et al., 2016] clip per-sample or per-client contributions and add Gaussian noise. Secure aggregation [Bonawitz et al., 2017] hides individual updates; sketching reduces communication [Rothchild et al., 2020]. DisCo-CFL applies clipping, sketching and the Gaussian mechanism to the clustering signal itself.

Fairness, explanations and accountability. Fair FL usually equalises accuracy across clients [Li et al., 2020b, Mohri et al., 2019, Li et al., 2021b]; see Mehrabi et al. [2021] for a broader view of bias. In CFL, unfairness arises earlier, in the grouping decision: a small group can be absorbed by a larger one, and a small client can be isolated because its statistics are noisy. Explanations that let an affected party understand and contest an automated decision [Wachter et al., 2018, Doshi-Velez and Kim, 2017] have, to our knowledge, not been provided for client grouping. The survey likewise notes that clustering-specific evaluation is rare [Ben Ali et al., 2026, Sec. 4.5.3].

2.6 Positioning

Table 1 compares representative methods on the properties this paper targets. The entries for other methods summarise what their papers and the survey [Ben Ali et al., 2026, Tables 3 and 5] report; “–” means that we could not find the property claimed. SA-compat.: compatible with secure aggregation. DisCo-CFL is, to our knowledge, the first CFL method that combines (i) a similarity that is invariant to label and quantity skew by construction and calibrated across sample sizes and privacy noise, (ii) an automatic K from a threshold with a fixed meaning, (iii) recovery guarantees independent of group size together with a matching lower bound, machine-checked in Lean 4, and (iv) explanations, certificates and an audit trail of the grouping decision.

Table 1: Positioning of DisCo-CFL. S/C/M: server-side, client-side, metadata-based. Shared: what the server clusters on. K : fixed (given), threshold (implicit in another hyperparameter) or calibrated (fixed-meaning threshold). LS-inv.: invariant to label skew by design. QS: explicit treatment of quantity skew. Priv.: formal privacy mechanism for the clustering signal. Expl./Cert.: explanations or certificates of the grouping. Theory: main guarantee.

Method	Cat.	Shared	K	LS-inv.	QS	Priv.	Expl.	Cert.	Theory
CFL/MTCFL [Sattler et al., 2021]	S	updates	threshold	–	–	SA-compat.	–	–	bipartition exists
FL+HC [Briggs et al., 2020]	S	updates	threshold	–	–	–	–	–	–
FlexCFL [Duan et al., 2021]	S	updates (SVD)	fixed	–	–	–	–	–	–
FeSEM [Long et al., 2023]	S	weights	fixed	–	–	–	–	–	–
StoCFL [Zeng et al., 2025]	S	frozen-model grads	threshold	–	–	–	–	–	–
IFCA [Ghosh et al., 2020]	C	losses (local)	fixed	–	–	–	–	–	convergence
FedEM [Marfoq et al., 2021]	C	mixture weights	fixed	–	–	–	–	–	convergence
FedRC [Guo et al., 2024]	C	losses (local)	fixed	–	–	–	–	–	–
k-FED [Dennis et al., 2021]	M	data centroids	fixed	–	–	–	–	–	one-shot recovery
PACFL [Vahidian et al., 2023]	M	data subspaces	threshold	–	–	–	–	–	–
pp-CFL [Luo et al., 2024]	M	noisy label dists.	COPRA	–	–	DP	–	–	–
CORNFLQS [Ben Ali et al., 2025]	S+C	weights + losses	iterative	–	✓	–	–	–	–
FedCCFA [Chen et al., 2024]	S	class classifiers	DBSCAN	partial	–	–	–	–	–
FedDAG [Pramanik et al., 2026]	S+M	grads + class subspaces	selected	partial	weights	–	–	–	–
DisCo-CFL	S	class-cond. grad. sketches	calibrated	✓	✓	DP	✓	✓	recovery + lower bound (Lean)

3 Problem setting

There are N clients. Client i holds $D_i = \{(x_s, y_s)\}_{s=1}^{n_i}$ with labels in $[C]$ drawn from $\mathcal{P}_i(X, Y) = \mathcal{P}_i(Y) \mathcal{P}_i(X | Y)$. Let $S_i = \{c : n_i^c \geq m\}$ be the classes the client holds with at least m samples.

Definition 1 (Concept equivalence and conflicts). Clients i and j *conflict on class* $c \in S_i \cap S_j$ if $\mathcal{P}_i(X | Y = c) \neq \mathcal{P}_j(X | Y = c)$. They are *compatible* if they conflict on no shared class. Clients belong to K latent concept groups $G_1, \dots, G_K$: $\mathcal{P}_i(X | Y = c) = \mathcal{P}^{(k)}(X | Y = c)$ for $i \in G_k$.

The target of clustering is concept equivalence, not equality of joint distributions. If $\mathcal{P}_i(X | Y) = \mathcal{P}_j(X | Y)$, the Bayes classifiers of i and j differ only through the priors $\mathcal{P}_i(Y), \mathcal{P}_j(Y)$, which a shared model can absorb by prior (logit) correction [Lipton et al., 2018, Menon et al., 2021]. Splitting such clients only reduces the data available to each model. Target skew and quantity skew therefore should *not* create clusters, while concept shift on features and on targets should. In the survey’s taxonomy [Ben Ali et al., 2026], a rotation of the inputs changes $\mathcal{P}(X | Y = c)$ for all classes, and a label swap $a \leftrightarrow b$ changes it for classes a and b .

4 DisCo-CFL

4.1 Class-conditional gradient signatures (client side)

After T_0 warm-up rounds of FedAvg the server broadcasts a reference model w and the seed of a count-sketch $\Phi : \mathbb{R}^d \rightarrow \mathbb{R}^p$ [Charikar et al., 2002, Weinberger et al., 2009] ($p = 1024$). For a sample (x, c) let $g(x, c) = \Phi \nabla_w \ell(w; x, c) \in \mathbb{R}^p$. For every class $c \in S_i$ the client randomly splits its class- c samples into halves A_i^c, B_i^c of size h_i^c and sends

$$a_i^c = \frac{1}{h_i^c} \sum_{s \in A_i^c} g(x_s, c), \quad b_i^c = \frac{1}{h_i^c} \sum_{s \in B_i^c} g(x_s, c).$$

Per-sample sketched gradients are clipped to a shared norm C_{clip} (0.5 in all experiments) before averaging. Per-sample gradients at a trained reference model are extremely heavy-tailed: in our Fashion-MNIST runs the median norm is 0.08 and the 99th percentile about 23, so a handful of hard examples would otherwise dominate a class signature. The expected clipped gradient is still a functional of $\mathcal{P}_i(X | Y = c)$, so invariance is preserved, and clipping makes per-sample signatures bounded, hence sub-Gaussian as required by Assumption 1. The upload happens *once* and costs $2p|S_i| \leq 2pC$ floats, independent of the model size d . For our small CNN ($d \approx 4.7 \times 10^4$) this is comparable to one model upload; for realistic models with $d \gg 2pC$ it is negligible. For differential privacy (Proposition 2), Gaussian noise is added to each clipped half-mean.

4.2 Disattenuated similarity (server side)

Let $m_i^c = (a_i^c + b_i^c)/2$. The split-half reliability and its Spearman–Brown extension to the full mean are

$$r_i^c = \cos(a_i^c, b_i^c), \quad R_i^c = \frac{2r_i^c}{1 + r_i^c} \quad (R_i^c := 0 \text{ if } r_i^c \leq 0). \quad (1)$$

The calibrated similarity of i and j on class c is

$$\hat{\rho}_{ij}^c = \text{clip}_{[-1,1]} \left(\frac{\cos(m_i^c, m_j^c)}{\sqrt{R_i^c R_j^c}} \right), \quad W_{ij}^c = R_i^c R_j^c, \quad (2)$$

Algorithm 1 DisCo-CFL

- 1: Warm-up: T_0 rounds of FedAvg on all clients $\rightarrow$ reference model w ; broadcast w and sketch seed
 - 2: **for** each client i **in parallel** **do** $\triangleright$ once
 - 3: **for** $c \in S_i$ **do** split class- c samples into halves, send (a_i^c, b_i^c) (per-sample clipped; noised if DP)
 - 4: **end for**
 - 5: **end for**
 - 6: Server: R_i^c by (1); $\hat{\rho}_{ij}^c, W_{ij}^c$ by (2)
 - 7: Clusters $\leftarrow$ singletons of clients with enough evidence
 - 8: **while** $\exists A, B$ with $L(A, B) \geq \tau$ **do** merge the eligible pair with the largest $\sum_c \text{den}_{AB}^c$; log it
 - 9: **end while**
 - 10: Attach low-evidence clients and non-significant singletons; flag anomalies
 - 11: Output clusters, certificates, explanations, audit log; run FedAvg within each cluster from w
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where W_{ij}^c is the evidence weight. In the population version of (2) the attenuation factors cancel exactly (Lemma 1), so $\hat{\rho}$ targets $\rho_{ij}^c = \cos(\mu_i^c, \mu_j^c)$, the cosine between expected class-conditional gradients, whatever the sample sizes and noise levels.

4.3 Threshold clustering with bottleneck linkage

For two sets of clients A, B and a class c , let $\text{num}_{AB}^c = \sum_{i \in A, j \in B} W_{ij}^c \hat{\rho}_{ij}^c$ and $\text{den}_{AB}^c = \sum W_{ij}^c$. The linkage is

$$L(A, B) = \min_c \frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} \quad \text{if } \sum_c \text{den}_{AB}^c \geq e_{\min}, \text{ else } -\infty. \quad (3)$$

The *minimum* over classes encodes the definition of concept equivalence: two groups share a concept only if every class-conditional gradient agrees, so a label swap affecting two of ten classes is not averaged away. The shrinkage λ pulls classes with little evidence towards 1 (“no evidence of a difference”). Starting from singletons, the server repeatedly merges, among the pairs with $L \geq \tau$, the pair with the *largest shared evidence*. Well-supported groups therefore form first, and clients that carry little information on the discriminative classes join them afterwards instead of bridging different groups. Merging stops when no pair reaches τ . Because the scale is calibrated, τ has a fixed meaning: we use $\tau = 0.8$ in *all* experiments, with $\lambda = 0.5$. Clients with total reliability $\sum_c R_i^c < 0.3$ are *low-evidence*: they are excluded from shaping clusters and attached to their most similar cluster afterwards. A singleton cluster is attached to its closest cluster unless the evidence *significantly* separates them (upper confidence bound below τ), in which case it is kept and flagged as *anomalous*. Each cluster then runs FedAvg from w . Algorithm 1 summarises the procedure.

4.4 Inference: prior-corrected cluster models

Clusters are built so that their members share class-conditionals but may have very different label proportions. By Bayes’ rule, the posterior of client i is obtained from that of its cluster k by re-weighting the prior (Proposition 4). Client i therefore predicts with

$$\hat{y}(x) = \arg \max_y [f_k(x)_y + \log \hat{p}_i(y) - \log \hat{p}_k(y)], \quad (4)$$

where f_k are the logits of the cluster model, $\hat{p}_i$ is the client’s own (smoothed) label distribution, and $\hat{p}_k$ is the label distribution of the cluster’s training data. The latter is a sum of label counts

that the server can obtain by secure aggregation, so no individual histogram is revealed. This correction [Lipton et al., 2018, Menon et al., 2021] costs nothing at training time. It is exactly what concept-level clustering calls for: label skew is handled by the prior, not by more clusters. In the experiments we apply the same correction to *every* method, for a fair comparison.

4.5 Transparency and accountability outputs

- **Class-level explanations.** For every pair of clusters the server reports the pooled per-class similarity and the set of *divergent classes* $\hat{D}_{AB} = \{c : \rho_{AB}^c < \tau\}$.
- **Heterogeneity diagnosis.** If $\hat{D}_{AB}$ contains every shared class, the server reports a global (feature-level) concept shift; if it is a strict subset, a class-specific shift on the listed classes; if $\hat{K} = 1$, it reports that the heterogeneity is explained by target and quantity skew only.
- **Assignment certificates.** Each client receives its bottleneck similarity to its own cluster and to the closest other cluster, the bottleneck classes, and standard errors from the spread over member pairs. A certificate is *confident* if $s_{\text{own}} - z \text{se} \geq \tau$ and $s_{\text{other}} + z \text{se} < \tau$ ($z = 2$); otherwise it is *uncertain* or *low-evidence*.
- **Audit log** of every merge (members, linkage, bottleneck class, evidence) and of every attachment or anomaly decision, so that each grouping can be traced and contested.
- **Newcomers** compute their signature at the same frozen w and join the cluster with the highest bottleneck similarity if it reaches τ . Otherwise they are reported as a *novel concept*.

5 Theory

Proofs are in Appendix A; their deterministic parts are formalised in Lean 4 (Appendix B). Write $\mu_i^c = \mathbb{E}_{x \sim \mathcal{P}_i(X|Y=c)}[g(x, c)]$ and $a_i^c = \|\mu_i^c\|^2$.

Proposition 1 (Invariance). μ_i^c is a functional of $\mathcal{P}_i(X | Y = c)$ only. Consequently, for compatible clients $\rho_{ij}^c = 1$ on every shared class, whatever their label proportions $\mathcal{P}_i(Y), \mathcal{P}_j(Y)$ and sample sizes n_i, n_j .

Assumption 1 (Noise). Per-sample signatures are $g = \mu_i^c + \Sigma^{1/2}z$ with independent z having independent, zero-mean, unit-variance, K_0 -sub-Gaussian coordinates. Let $\sigma^2 = \text{tr } \Sigma$, and define the effective ranks $r_s = \text{tr } \Sigma / \|\Sigma\|_{\text{op}}$ and $r_e = (\text{tr } \Sigma)^2 / \|\Sigma\|_F^2$.

Assumption 2 (Identifiability). If i, j conflict on c , then $\rho_{ij}^c \leq 1 - \Delta$ for some $\Delta \in (0, 1]$.

For a pair and a class let $h = \min(h_i^c, h_j^c)$, $a = \min(a_i^c, a_j^c)$, and let $\kappa = \sigma^2/(ah)$ be the inverse signal-to-noise ratio of a half-mean.

Lemma 1 (Exact calibration at the population level). Replacing each inner product and squared norm in (1)–(2) by its expectation gives exactly ρ_{ij}^c .

Theorem 1 (Concentration). Under Assumption 1, let $\kappa \leq 1$ and $\delta \in (0, 1)$. There are constants c_0, C_1 depending only on K_0 such that, if

$$\eta := \left(\sqrt{\kappa/r_s} + \kappa/\sqrt{r_e} \right) \sqrt{\log(12/\delta)} \leq c_0,$$

then with probability at least $1 - \delta$

$$|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq C_1 \eta, \quad |R_i^c - R_i^{c,\star}| \leq C_1 \eta, \quad R_i^{c,\star} := \frac{a_i^c}{a_i^c + \sigma^2/(2h_i^c)}.$$

In contrast, the raw cosine $\cos(m_i^c, m_j^c)$ converges to $\rho_{ij}^c \sqrt{R_i^{c,*} R_j^{c,*}}$. It is biased towards 0 by a factor that depends on the sample sizes.

Theorem 2 (Safety, maximality, recovery). *Let $\gamma = \min(1-\tau, \tau-1+\Delta) > 0$ and $w_0 := 2\lambda(1-\tau)/\gamma$. Suppose that all estimates satisfy $|\hat{\rho}_{ij}^c - \rho_{ij}^c| \leq \gamma/2$ (event $\mathcal{E}$) and that every pair conflicting on a class c has evidence $W_{ij}^c > w_0$. Then the merging phase of Algorithm 1, in any merge order, satisfies:*

- (a) Safety: no cluster ever contains two conflicting clients.
- (b) Maximality: any two output clusters with shared evidence $\geq e_{\min}$ contain a conflicting pair, so no two output clusters can be merged without mixing concepts.
- (c) Exact recovery: if every client holds every class on which some groups differ, and every pair of clients shares evidence $\geq e_{\min}$, the output partition equals $\{G_1, \dots, G_K\}$; in particular $\hat{K} = K$.

Corollary 1 (Sample complexity and minority fairness). *Under Assumptions 1–2, let $\gamma \leq 1/3$ and $\lambda < \gamma/(8(1-\tau))$. The conclusions of Theorem 2 hold with probability at least $1 - \delta$ as soon as every reported half size satisfies*

$$h \geq h^* := C_2 \frac{\sigma^2}{a} \max \left\{ \frac{L}{r_s \gamma^2}, \frac{\sqrt{L}}{\sqrt{r_e} \gamma}, 1 \right\}, \quad L := \log \frac{12N^2 C}{\delta}. \quad (5)$$

h^* does not depend on the sizes of the groups $|G_k|$, on the number of clients per group, or on the other clients' dataset sizes (apart from $\log N$). A concept group made of a single client is recovered under the same condition as a group of a hundred.

Theorem 3 (Minimax lower bound). *Let $\xi \sim \mathcal{N}(0, \Sigma)$ and $\|\mu\|^2 = a$. For any test that decides from n per-sample signatures of client j , even when μ_i is known exactly, whether $\rho_{ij} = 1$ or $\rho_{ij} = 1 - \Delta$, the worst-case error probability is at least*

$$\frac{1}{4} \exp(-n a \Delta r_s / \sigma^2).$$

Error at most δ therefore requires

$$n \geq \frac{\sigma^2}{a r_s \Delta} \log \frac{1}{4\delta}.$$

With τ in the middle of the admissible interval ($\gamma = \Delta/2$), the leading term of (5) is

$$O\left(\frac{\sigma^2 \log(N^2 C / \delta)}{a r_s \Delta^2}\right).$$

The upper and lower bounds therefore agree in their dependence on noise σ^2 , signal a and effective rank r_s , and differ by a factor $1/\Delta$ and a $\log N$ term.

Proposition 2 (Differential privacy and its cost). *Suppose each released half-mean is*

$$\frac{1}{h} \sum_s \text{clip}_{C_{\text{clip}}}(g_s) + \xi_{\text{dp}}, \quad \xi_{\text{dp}} \sim \mathcal{N}(0, (2\sigma_{\text{dp}} C_{\text{clip}}/h)^2 I_p).$$

(i) Every sample enters exactly one released vector, whose replace-one sensitivity is $2C_{\text{clip}}/h$. The signature is therefore (ε, δ) -DP with respect to replacing one record by another of the same label, with (ε, δ) given by the analytic Gaussian mechanism [Balle and Wang, 2018] at noise multiplier σ_{dp} (parallel composition). The class support S_i is released in the clear. (ii) The DP noise is

independent of sampling noise. Lemma 1 therefore still holds, and Theorem 1 applies with the half-mean covariance $\Sigma/h + (2\sigma_{\text{dp}}C_{\text{clip}}/h)^2I_p$. The recovery guarantee then requires, in addition to (5),

$$h \gtrsim \sigma_{\text{dp}}C_{\text{clip}} \left(\frac{\sqrt{\mathbb{L}}}{\sqrt{a}\gamma} + \frac{p^{1/4}\mathbb{L}^{1/4}}{\sqrt{a}\gamma} \right),$$

so the per-class sample size grows linearly in $\sigma_{\text{dp}} \propto 1/\varepsilon$. (iii) Without disattenuation, the raw cosine tends to $\rho \sqrt{R_i R_j}$ with

$$R \approx \frac{a}{a + 2p\sigma_{\text{dp}}^2 C_{\text{clip}}^2 / h^2}.$$

Unless $h \gg \sqrt{p}\sigma_{\text{dp}}C_{\text{clip}}/\sqrt{a}$, same-concept clients look unrelated.

Proposition 3 (Explanation fidelity). *On $\mathcal{E}$, if two output clusters A, B have pooled evidence $\text{den}_{AB}^c > w_0$ on a class c , then $c \in \hat{D}_{AB}$ if and only if $\mathcal{P}_A(X | Y = c) \neq \mathcal{P}_B(X | Y = c)$. The reported divergent classes are exactly the classes on which the two concepts differ.*

Proposition 4 (Prior correction is Bayes-optimal within a concept cluster). *Let cluster k consist of concept-equivalent clients with pooled distribution $\mathcal{P}_k(X, Y) = \mathcal{P}_k(Y)\mathcal{P}(X | Y)$, and let $q_k(y | x)$ be its Bayes posterior. For every member i , $\mathcal{P}_i(y | x) \propto q_k(y | x) \mathcal{P}_i(y) / \mathcal{P}_k(y)$. Hence (4) with the true priors and $f_k = \log q_k$ is the Bayes-optimal classifier of client i . No clustering finer than concept equivalence can reduce the Bayes risk of any member.*

Remark 1 (Scope of the guarantees). Theorem 2 covers the merging phase. The attachment of low-evidence clients and of non-significant singletons is a fairness heuristic: it avoids training a client alone because of noise, may violate safety when evidence is weak, and is logged in the audit trail. Corollary 1 asks for a small λ so that a single pair of clients carries enough evidence. Our default $\lambda = 0.5$ is larger. It makes the method more conservative about splitting on thin evidence, and merges quickly happen between clusters whose pooled evidence far exceeds w_0 . Section 7 reports the sensitivity to λ , including values that satisfy the corollary.

6 Fairness, accountability, transparency and privacy

Fairness. DisCo-CFL is fair in the clustering decision along two axes. *Group size:* Corollary 1 makes recovery independent of cluster size. By contrast, objectives of the k -means type (FL+HC, IFCA) favour large clusters, and IFCA’s analysis depends on the smallest cluster fraction [Ghosh et al., 2020]. *Data size:* small clients are not pushed away by attenuation (Theorem 1), and clients that are merely noisy are attached rather than isolated. We measure worst-group accuracy, the accuracy of the worst 10% of clients, Jain’s index [Jain et al., 1984], and minority recall and purity.

Accountability. Certificates, audit log and anomaly flags make every grouping decision traceable and contestable. We measure the *calibration* of certificates: how often a confident certificate belongs to a client that shares no conflict with its cluster.

Transparency. The explanations and diagnosis tell participants *why* they were separated (“you differ from cluster 2 on classes 3 and 5”), with a fidelity guarantee (Proposition 3). We measure precision, recall and exact match of the divergent-class sets, and the accuracy of the diagnosis.

Privacy. Proposition 2 gives record-level DP for the only extra information released. The signatures are low-dimensional sketches ($p \ll d$) and are uploaded once. Section 7 studies the resulting privacy–utility trade-off.

7 Experiments

7.1 Setup

Datasets. We use four image datasets of increasing difficulty: MNIST [LeCun et al., 1998], Fashion-MNIST [Xiao et al., 2017] (the main benchmark), SVHN [Netzer et al., 2011] (natural photographs of house numbers) and CIFAR-10 [Krizhevsky, 2009] (natural object images). Federations are built from them following the non-IID taxonomy used in the survey [Ben Ali et al., 2026], which lists image classification with rotations and label swaps as the standard CFL benchmark [Ben Ali et al., 2026, Table 6].

Heterogeneity. Each client belongs to a latent concept group. *Concept shift on features* is a rotation of the inputs (by $0^\circ, 90^\circ, 180^\circ, 270^\circ$, or by $0, \theta, 2\theta, 3\theta$ in the strength sweep). *Concept shift on targets* is a label permutation that swaps two pairs of classes per group. On top of this, every client has *target skew* (Dirichlet label proportions, $\alpha = 0.3$, or $\alpha = 0.1$ in the label-only scenario) and, where indicated, *quantity skew* (log-normal dataset sizes with $\sigma = 1$, clipped to $[30, 3200]$, mean 400). Table 2 lists all scenarios. Each client has a local test set drawn with its own label proportions and transformation. We also evaluate every model on a class-balanced test set of its client’s concept (*concept accuracy*), which measures generalisation beyond the local label mix.

Table 2: Scenarios. N : clients; groups: number of concept groups; QS: quantity skew; part.: fraction of clients per round.

Scenario	Dataset(s)	Concept groups	N	QS	Other
Rotation	FMNIST, MNIST, SVHN, CIFAR-10	4 rotations (90° steps)	40	–	
Label swap	FMNIST, MNIST, SVHN, CIFAR-10	4 labelings (2 swapped pairs)	40	–	
Rotation + QS	FMNIST	4 rotations	40	✓	
Mixed + QS	FMNIST	2 rotations $\times$ 2 labelings	40	✓	
Label skew only	FMNIST	1 (true $K = 1$)	40	✓	$\alpha = 0.1$
Minority groups	FMNIST	4 rotations, 25/9/4/2 clients	40	–	
Strength θ	FMNIST	4 rotations, $\theta \in \{15^\circ, 30^\circ, 45^\circ\}$	40	–	
Groups K	FMNIST	$K \in \{2, 4, 8\}$ labelings	40	–	
Partial participation	FMNIST	rotation / label swap	40	–	part. = 0.3
Scale	FMNIST	4 rotations	200	–	200 samples/client

Training. All methods use the same CNN (two convolutional and two fully connected layers; $d = 46,730$ parameters for 28×28 inputs and 65,962 for 32×32 colour images), SGD with learning rate 0.05, batch size 32, one local epoch per round and 50 communication rounds. Methods with a warm-up (DisCo-CFL, FL+HC, FeSEM, Oracle) spend 10 of the 50 rounds on FedAvg. *Local* trains each client alone for 50 epochs. Every configuration is run with three seeds, which change the federation, the initialisation and the sketch: 618 training runs in total, plus the clustering-only studies.

Baselines. We compare with nine methods that cover all three families of the taxonomy and the main alternatives to clustering:

- *Single model / personalisation:* FedAvg [McMahan et al., 2017]; Local; FedAvg-FT (FedAvg followed by one epoch of local fine-tuning, a standard personalisation baseline).
- *Client-side:* IFCA [Ghosh et al., 2020].
- *Server-side:* CFL/MTCFL [Sattler et al., 2021] (recursive cosine bipartitioning, triggered when the norm of the aggregated update falls below $\varepsilon_1 = 0.9$ times the mean client-update norm); FL+HC [Briggs et al., 2020] (Ward clustering of local updates after the warm-up); FeSEM [Long et al., 2023] (EM-like assignment to the nearest of K centres in parameter space, with the same warm-up and a k -means++ initialisation on the clients’ local models).
- *Metadata-based:* PACFL [Vahidian et al., 2023] (principal angles between client data subspaces, $p = 3$).
- *Oracle:* FedAvg inside the ground-truth groups (an upper reference).

IFCA, FL+HC, FeSEM and PACFL are *given the true number of groups K* , which favours them. DisCo-CFL and MTCFL find K on their own. The MTCFL thresholds and the FeSEM initialisation were selected on the same pilot run as DisCo-CFL. Without them, MTCFL never splits and FeSEM collapses to a single cluster.

DisCo-CFL. Sketch dimension $p = 1024$, at most 200 samples per class for the signature, classes with fewer than 4 samples not reported, $\tau = 0.8$, $\lambda = 0.5$, $e_{\min} = 0.05$, low-evidence threshold 0.3, $z = 2$, per-sample clipping $C_{\text{clip}} = 0.5$. These values, and the design choices of bottleneck linkage and evidence-ordered merging, were fixed during pilot runs on seed 0 of the Rotation, Label swap, Mixed + QS and label-only configurations (and seed 1 of Label swap with QS), using clustering quality only. They are kept identical for all scenarios, seeds and datasets. Seeds 1 and 2, the sweeps, and the MNIST, SVHN and CIFAR-10 results were not used for design. Per-sample clipping (C_{clip} chosen on seed 0 only, from $\{0.5, 1, 2\}$ in all scenarios and additionally $\{0.1, 0.25\}$ in the label-only scenario) was added after a first sensitivity study showed over-splitting caused by heavy-tailed per-sample gradients. The run without clipping is reported as an ablation. On the colour datasets we also report DisCo-T30, which uses 30 warm-up rounds (Section 7.3). Section 7.9 reports the sensitivity to τ and λ .

Metrics. *Performance:* mean local test accuracy (Acc), with the prior correction of Section 4.4 applied identically to every method, and concept accuracy. *Fairness:* accuracy of the worst 10% of clients (Acc@10%), accuracy of the worst concept group (Worst-grp), Jain’s index. *Clustering:* ARI with respect to the groups, number of clusters K , and the *conflict-merge rate*: the fraction of conflicting client pairs (Definition 1) that end up in the same cluster. ARI penalises the legitimate placement of compatible clients (e.g. a client of a label-swap group that holds none of the swapped classes), whereas the conflict-merge rate measures exactly the harmful errors. *Statistics:* paired one-sided Wilcoxon signed-rank tests of DisCo-CFL against each baseline over all (scenario, seed) pairs.

7.2 Main results: performance, fairness and clustering quality

Table 3 reports all metrics on Fashion-MNIST for the ten methods. Accuracy is *with* the prior correction (4), applied identically to every method; the uncorrected mean accuracy is shown in the first column.

Table 3: Fashion-MNIST, 40 clients, 3 seeds. Acc: mean local accuracy (with prior correction, $\pm$ std over seeds). Acc (raw): without correction. Acc@10%: mean accuracy of the worst 10% of clients. Worst-grp: accuracy of the worst concept group. Concept: accuracy on the class-balanced test set of the client’s concept. Conflict%: fraction of conflicting client pairs merged into one cluster (lower is better). IFCA, FeSEM, FL+HC and PACFL receive the true K ; DisCo-CFL finds it. Oracle uses the ground-truth groups. Best non-oracle values in bold.

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>Rotation</i>									
FedAvg	76.6	86.0 $\pm$ 0.5	76.5	82.9	75.7	0.99	0.00	1.0	100.0
Local	87.8	87.8 $\pm$ 0.7	80.8	86.9	60.7	1.00	0.00	40.0	0.0
FedAvg-FT	88.8	88.8 $\pm$ 0.5	81.5	88.0	77.9	1.00	0.00	40.0	0.0
IFCA	83.5	89.2 $\pm$ 1.0	80.9	86.7	77.0	1.00	0.41	4.0	22.0
FeSEM	83.0	89.1 $\pm$ 0.3	80.8	87.8	80.6	1.00	0.67	4.0	10.4
MTCFL	89.4	90.2 $\pm$ 0.6	83.6	89.1	73.9	1.00	0.15	34.7	0.0
FL+HC	78.5	86.4 $\pm$ 0.4	76.9	83.6	74.9	0.99	0.01	4.0	85.0
PACFL	82.8	89.5 $\pm$ 0.8	81.8	87.8	79.4	1.00	0.48	4.0	21.0
DisCo	84.5	90.5$\pm$0.4	84.6	89.5	83.9	1.00	1.00	4.0	0.0
Oracle	84.5	90.5 $\pm$ 0.4	84.6	89.5	83.9	1.00	1.00	4.0	0.0
<i>Label swap</i>									
FedAvg	54.2	70.1 $\pm$ 1.4	47.2	58.6	55.7	0.94	0.00	1.0	100.0
Local	88.7	88.7 $\pm$ 0.7	80.1	87.0	64.0	1.00	0.00	40.0	0.0
FedAvg-FT	90.6	90.6 $\pm$ 0.5	84.1	89.7	80.9	1.00	0.00	40.0	0.0
IFCA	76.9	85.6 $\pm$ 1.6	72.6	79.4	71.1	0.99	0.45	4.0	17.1
FeSEM	80.4	87.5 $\pm$ 5.0	76.3	83.0	79.8	0.99	0.87	4.0	5.7
MTCFL	88.2	90.4 $\pm$ 0.6	84.3	87.8	76.1	0.99	0.30	29.0	2.8
FL+HC	58.4	71.5 $\pm$ 1.1	47.0	61.1	59.1	0.93	0.01	4.0	85.7
PACFL	57.9	74.8 $\pm$ 2.6	50.7	67.8	53.6	0.94	-0.01	4.0	34.9
DisCo	85.4	90.8$\pm$0.3	84.6	89.2	84.9	1.00	1.00	4.0	0.0
Oracle	85.4	90.8 $\pm$ 0.3	84.6	89.2	84.9	1.00	1.00	4.0	0.0
<i>Rotation + QS</i>									
FedAvg	75.9	85.8 $\pm$ 0.3	78.4	82.6	74.7	1.00	0.00	1.0	100.0
Local	87.9	87.9 $\pm$ 1.2	80.4	84.9	61.9	1.00	0.00	40.0	0.0
FedAvg-FT	87.6	87.6 $\pm$ 0.7	80.8	84.8	75.4	0.99	0.00	40.0	0.0
IFCA	82.3	88.5 $\pm$ 0.6	81.7	86.1	77.0	1.00	0.44	3.3	24.3
FeSEM	81.2	88.3 $\pm$ 0.6	81.6	85.5	78.4	0.99	0.66	4.0	11.6
MTCFL	85.4	88.8 $\pm$ 0.7	82.7	86.7	73.5	1.00	0.32	20.7	15.6
FL+HC	76.9	85.3 $\pm$ 1.0	76.1	80.2	73.5	0.99	0.01	4.0	85.0
PACFL	80.1	88.0 $\pm$ 1.8	79.7	84.3	76.5	0.99	0.44	4.0	20.8
DisCo	83.3	90.1$\pm$0.2	84.5	88.7	82.1	1.00	1.00	4.0	0.0
Oracle	83.3	90.1 $\pm$ 0.2	84.5	88.7	82.1	1.00	1.00	4.0	0.0

Table 3 (continued).

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>Mixed + QS</i>									
FedAvg	70.2	80.8±0.9	66.3	75.1	66.6	0.98	0.00	1.0	100.0
Local	88.9	88.9±1.3	82.7	86.6	64.1	1.00	0.00	40.0	0.0
FedAvg-FT	88.6	88.6±0.9	82.9	85.7	76.5	0.99	0.00	40.0	0.0
IFCA	82.5	89.0±0.3	82.5	87.5	75.4	1.00	0.34	3.7	25.5
FeSEM	81.1	88.1±1.3	79.4	85.4	77.9	0.99	0.67	4.0	8.3
MTCFL	78.4	84.8±1.8	70.0	76.8	70.2	0.98	0.24	16.3	29.3
FL+HC	72.0	81.8±0.3	68.6	77.4	67.1	0.98	0.01	4.0	85.4
PACFL	71.5	83.3±0.4	68.3	79.3	66.4	0.98	0.20	4.0	21.8
DisCo	84.5	90.6±0.3	84.3	88.8	82.5	1.00	0.94	4.0	0.6
Oracle	84.2	90.6±0.3	84.6	88.7	83.4	1.00	1.00	4.0	0.0
<i>Label skew only</i>									
FedAvg	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
Local	94.6	94.6±0.2	87.7	94.6	51.0	1.00	0.00	40.0	0.0
FedAvg-FT	94.2	94.2±0.3	90.1	94.2	80.7	1.00	0.00	40.0	0.0
IFCA	87.7	95.4±0.6	90.7	95.4	85.7	1.00	1.00	1.0	0.0
FeSEM	87.7	95.5±0.4	90.6	95.5	85.7	1.00	1.00	1.0	0.0
MTCFL	92.0	95.3±0.2	88.5	95.3	77.1	1.00	0.00	20.7	0.0
FL+HC	87.5	95.5±0.5	90.4	95.5	85.9	1.00	1.00	1.0	0.0
PACFL	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
DisCo	87.6	95.6±0.4	90.8	95.6	85.6	1.00	0.67	1.3	0.0
Oracle	87.6	95.5±0.4	90.9	95.5	85.8	1.00	1.00	1.0	0.0
<i>Minority groups</i>									
FedAvg	77.0	85.5±0.7	74.1	35.7	76.7	0.98	0.00	1.0	100.0
Local	88.5	88.5±0.5	81.0	85.2	62.7	1.00	0.00	40.0	0.0
FedAvg-FT	89.4	89.4±0.5	82.9	81.7	78.5	1.00	0.00	40.0	0.0
IFCA	83.6	89.4±1.7	81.2	67.3	78.1	0.99	0.57	3.7	22.8
FeSEM	84.0	89.8±0.7	82.0	79.4	81.8	1.00	0.62	4.0	5.7
MTCFL	82.0	87.3±3.1	78.0	51.9	76.7	0.98	0.04	11.3	66.7
FL+HC	80.4	87.7±0.8	78.0	74.5	78.0	0.99	0.22	4.0	74.8
PACFL	83.1	89.8±0.7	83.9	82.7	78.8	1.00	0.43	4.0	17.8
DisCo	85.3	90.8±0.6	85.3	86.5	84.1	1.00	0.98	4.3	0.0
Oracle	85.3	90.9±0.6	85.3	86.5	84.2	1.00	1.00	4.0	0.0

DisCo-CFL recovers the concept structure without being told K . In *Rotation*, *Label swap* and *Rotation + QS*, DisCo-CFL finds exactly the ground-truth partition in all seeds (ARI = 1) and matches the Oracle on every metric. In *Mixed + QS*, where two groups differ only on 4 of the 10 classes, it reaches ARI 0.94 with 0.6% conflicting pairs merged. In *Minority groups* (25/9/4/2 clients) it reaches ARI 0.98 with no conflicts. The baselines that are given the true K merge 6–86% of the conflicting pairs: FeSEM 6–12%, IFCA 17–26%, PACFL 18–35% and FL+HC 75–86%. CFL/MTCFL has few conflicts only in the scenarios where it fragments the federation into 16–35 clusters. FL+HC groups clients by label proportions instead of concepts, PACFL by the pixel subspace (which a label swap leaves unchanged), IFCA suffers from initialisation, and FeSEM, the strongest clustering baseline, confuses nearby groups in parameter space.

Label skew does not create clusters. With a single concept and strong target and quantity skew (*Label skew only*), DisCo-CFL returns $K = 1$ in two seeds. In the third it isolates one client whose signature for one class disagrees with all others, and flags it as *anomalous* in the audit log. Accuracy equals FedAvg’s. CFL/MTCFL, in contrast, splits the federation into about 20 clusters, and its concept accuracy drops from 85.8% to 77.1%.

Performance. With the prior correction, DisCo-CFL has the best mean accuracy in every Fashion-MNIST scenario. Personalising and fragmenting methods (Local, FedAvg-FT, MTCFL with 16–35 clusters) obtain competitive *local* accuracy, because local test sets share the client’s skewed label mix. In the concept-shift scenarios their *concept* accuracy is 4–23 points lower: their models do not generalise to the rest of the client’s concept, e.g. to classes whose frequency will change. DisCo-CFL is best on concept accuracy in every concept-shift scenario (82.1–84.9%, against at most 81.8% for any baseline). The prior correction adds 5–17 points to every method that shares models across clients. It is what makes concept-level clustering competitive on local metrics, as Proposition 4 predicts.

Fairness. DisCo-CFL has the best accuracy on the worst 10% of clients in every scenario with concept shift, and the best worst-group accuracy in all of them except *Label swap*, where FedAvg-FT is 0.5 points higher (89.7% vs. 89.2%). The gap is largest for minority concepts. In *Minority groups*, the worst concept group reaches 86.5% with DisCo-CFL, against 35.7% with FedAvg, 51.9% with MTCFL, 67.3% with IFCA, 74.5% with FL+HC and 79.4% with FeSEM: the size-independence of Corollary 1 in practice. Table 4 isolates the clustering step. DisCo-CFL recovers minority groups of 1, 2, 3 and 5 clients with recall and purity 1.00 and no conflicting merges. FL+HC isolates the minority only by spending its fixed K on it while merging 81% of the conflicting pairs among the majority groups, and PACFL does not find the minority at all.

Table 4: Minority-group recovery (clustering step only; groups of 20/10/10/ m clients; 3 seeds).

m	DisCo-CFL			FL+HC (K given)		
	recall	ARI	conflict%	recall	ARI	conflict%
1	1.00	0.98	0.0	1.00	0.09	81.5
2	1.00	0.98	0.0	1.00	0.11	81.0
3	1.00	1.00	0.0	1.00	0.12	80.6
5	1.00	0.98	0.0	0.60	0.09	82.9

7.3 Other datasets: MNIST, SVHN and CIFAR-10

Table 6 repeats the two basic concept-shift scenarios on MNIST, SVHN and CIFAR-10 with the same hyperparameters.

MNIST and SVHN. On both datasets DisCo-CFL recovers the groups (ARI 0.97–1.00), never merges conflicting clients, and reaches the accuracy of the Oracle within 0.1 points. On SVHN, a harder dataset of natural photographs, its concept accuracy is 79.7% (rotation) and 82.5% (label swap), against at most 75.6% and 72.7% for any baseline. FeSEM is the strongest clustering baseline on these datasets (ARI 0.66–0.98), and FedAvg-FT the strongest non-clustering one. Both remain 0.7–11.6 points below DisCo-CFL in concept accuracy. In the MNIST rotation scenario, two seeds leave one or two clients in singleton clusters. These clients are flagged as anomalous (mean ARI 0.97, $K = 5.0$) and train alone; the mean accuracy still equals the Oracle’s.

CIFAR-10. CIFAR-10 is harder, and here DisCo-CFL is not the best method on every metric. With the default 10 warm-up rounds the reference model is only about 40% accurate. Its class-conditional gradients then depend little on the orientation of most natural-image classes: the mean between-group similarity is 0.83–0.90, and only vehicle classes separate clearly. The separation Δ of Assumption 2 is therefore small. For label swaps DisCo-CFL still reaches ARI 0.87 and a higher concept accuracy than every baseline (48.6%), but for rotations it reaches only ARI 0.38. The theory points to the remedy, a more informative reference model. With 30 warm-up rounds (DisCo-T30), ARI rises to 0.74 (rotation) and 0.92 (label swap), conflicting merges fall to 4.2% and 0.1%, and concept accuracy is 44.8% and 49.8%. That is the best of all non-oracle methods in both scenarios, tied with FeSEM on rotation. Table 5 gives the clustering-only trend. CFL/MTCFL has the highest *local* accuracy on CIFAR-10 (67.0% and 68.8%, above the Oracle’s 64.2% and 64.7%). It splits the federation into 33–35 clusters and personalises to each client’s label mix with the small CNN, and its concept accuracy is 7–8 points below DisCo-T30’s.

Table 5: CIFAR-10 rotation, clustering step only: effect of the warm-up length T_0 (3 seeds).

T_0	ARI	K	conflict%
10	0.37	3.7	24.8
20	0.64	5.0	8.9
30	0.74	4.7	4.2

7.4 Robustness sweeps

Table 7 varies one factor at a time on Fashion-MNIST: the strength of the concept shift, the number of groups, the fraction of participating clients and the number of clients. Baselines are given the true K where they need it.

Strength of the shift. With rotations of $0, \theta, 2\theta, 3\theta$ degrees, DisCo-CFL separates the groups perfectly from $\theta = 45^\circ$ on and almost perfectly at 30° (ARI 0.93). At 15° neighbouring concepts are so close that their class-conditional gradients agree above the threshold, and DisCo-CFL merges them (ARI 0.42). This is the behaviour predicted by the separation assumption: the method groups what it cannot tell apart. It costs little, because nearby concepts can share a model: accuracy is 90.0% against 90.5% for the Oracle, and 0.2 points below FeSEM, which is given $K = 4$. For every

Table 6: MNIST, SVHN and CIFAR-10 (40 clients, 3 seeds; same settings and columns as Table 3). DisCo-T30: DisCo-CFL with 30 warm-up rounds.

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>MNIST rotation</i>									
FedAvg	91.8	94.7±0.6	91.1	92.4	90.6	1.00	0.00	1.0	100.0
Local	96.5	96.5±0.4	94.3	96.1	82.0	1.00	0.00	40.0	0.0
FedAvg-FT	97.2	97.2±0.1	95.5	97.0	94.6	1.00	0.00	40.0	0.0
IFCA	95.5	96.9±0.5	94.4	95.6	91.2	1.00	0.42	4.0	21.6
FeSEM	97.3	98.2±0.1	97.0	97.8	95.9	1.00	0.78	4.0	5.3
MTCFL	93.8	95.7±0.7	92.8	92.9	90.8	1.00	0.09	9.7	58.9
FL+HC	91.4	94.3±0.9	88.4	91.6	89.9	1.00	0.01	4.0	85.0
PACFL	95.1	96.8±0.3	94.5	95.7	91.6	1.00	0.45	4.0	15.9
DisCo	97.6	98.5±0.1	97.3	98.3	97.1	1.00	0.97	5.0	0.0
Oracle	97.6	98.5±0.1	97.3	98.3	97.2	1.00	1.00	4.0	0.0
<i>MNIST label swap</i>									
FedAvg	60.4	78.5±2.3	52.8	61.9	60.9	0.94	0.00	1.0	100.0
Local	97.1	97.1±0.2	95.0	96.4	86.0	1.00	0.00	40.0	0.0
FedAvg-FT	98.1	98.1±0.0	96.8	97.5	92.0	1.00	0.00	40.0	0.0
IFCA	90.9	95.8±0.6	92.4	90.4	87.9	1.00	0.60	3.7	13.3
FeSEM	97.4	98.4±0.4	97.6	97.4	97.1	1.00	0.98	4.0	0.6
MTCFL	85.7	91.7±3.2	82.2	85.0	78.7	0.98	0.24	18.0	24.2
FL+HC	66.0	80.8±2.0	56.6	69.6	65.9	0.95	0.01	4.0	85.7
PACFL	67.2	83.3±2.1	62.2	75.5	63.4	0.97	0.00	4.0	30.0
DisCo	97.9	98.7±0.1	97.7	98.5	97.8	1.00	1.00	4.0	0.0
Oracle	97.9	98.7±0.1	97.7	98.5	97.8	1.00	1.00	4.0	0.0
<i>SVHN rotation</i>									
FedAvg	63.3	72.4±1.2	58.1	64.5	63.4	0.98	0.00	1.0	100.0
Local	77.5	77.5±1.0	69.0	75.6	43.9	0.99	0.00	40.0	0.0
FedAvg-FT	81.9	81.9±0.2	77.1	81.0	69.5	1.00	0.00	40.0	0.0
IFCA	74.8	80.2±1.6	71.5	76.9	65.9	0.99	0.26	3.7	25.8
FeSEM	79.8	84.1±0.3	79.0	82.3	75.6	1.00	0.66	4.0	9.4
MTCFL	82.4	84.0±0.2	79.9	83.0	66.9	1.00	0.27	24.7	10.6
FL+HC	65.3	72.9±1.4	57.6	64.9	64.4	0.98	0.01	4.0	85.0
PACFL	70.2	78.1±1.3	67.7	75.9	67.5	0.99	0.39	4.0	16.9
DisCo	81.3	85.4±0.2	80.4	83.9	79.7	1.00	0.99	4.3	0.0
DisCo-T30	80.7	84.6±0.3	80.4	83.5	78.6	1.00	0.93	5.7	0.0
Oracle	81.5	85.5±0.3	80.5	84.3	80.0	1.00	1.00	4.0	0.0

Table 6 (continued).

Method	Acc (raw)	Acc	Acc@10%	Worst-grp	Concept	Jain	ARI	K	Conflict%
<i>SVHN label swap</i>									
FedAvg	52.6	64.5±3.7	44.9	54.5	50.9	0.94	0.00	1.0	100.0
Local	81.0	81.0±1.0	75.0	79.7	50.4	1.00	0.00	40.0	0.0
FedAvg-FT	84.9	84.9±0.1	79.8	84.3	72.7	1.00	0.00	40.0	0.0
IFCA	71.4	79.0±2.6	65.2	74.9	64.1	0.98	0.37	4.0	21.2
FeSEM	73.7	80.5±6.0	71.9	76.5	70.9	0.98	0.67	4.0	12.8
MTCFL	80.0	83.1±3.4	74.0	79.4	66.5	0.99	0.26	25.0	10.7
FL+HC	56.2	66.6±3.2	45.8	57.7	53.4	0.94	0.01	4.0	84.7
PACFL	54.2	69.1±1.9	53.6	60.3	48.2	0.97	-0.01	4.0	28.9
DisCo	83.7	87.2±0.4	83.5	86.1	82.5	1.00	0.97	4.3	0.0
DisCo-T30	83.8	87.1±0.2	83.5	86.4	82.1	1.00	0.94	5.7	0.0
Oracle	83.6	87.1±0.4	83.5	86.0	82.6	1.00	1.00	4.0	0.0
<i>CIFAR-10 rotation</i>									
FedAvg	38.0	55.1±1.4	44.5	51.6	36.8	0.97	0.00	1.0	100.0
Local	56.2	56.2±3.6	42.7	52.1	21.0	0.96	0.00	40.0	0.0
FedAvg-FT	61.8	61.8±2.0	51.1	60.4	36.8	0.98	0.00	40.0	0.0
IFCA	45.9	56.5±1.6	44.1	53.4	32.0	0.97	-0.01	4.0	32.2
FeSEM	49.4	61.9±1.5	52.3	59.0	44.8	0.98	0.69	4.0	12.1
MTCFL	65.6	67.0±2.2	56.8	63.3	38.2	0.98	0.14	34.7	0.0
FL+HC	40.5	55.6±1.6	43.5	54.4	36.9	0.97	0.01	4.0	85.0
PACFL	46.2	58.6±2.4	48.1	54.8	37.3	0.98	0.37	4.0	21.2
DisCo	47.6	60.5±1.2	47.5	56.3	41.0	0.98	0.38	3.7	24.6
DisCo-T30	50.4	62.0±1.3	51.5	60.3	44.8	0.98	0.74	4.7	4.2
Oracle	52.0	64.2±1.7	54.3	61.6	47.4	0.99	1.00	4.0	0.0
<i>CIFAR-10 label swap</i>									
FedAvg	34.8	51.0±2.9	36.0	43.0	33.5	0.95	0.00	1.0	100.0
Local	56.4	56.4±3.7	42.8	51.7	21.8	0.96	0.00	40.0	0.0
FedAvg-FT	64.5	64.5±2.7	53.9	62.9	40.8	0.98	0.00	40.0	0.0
IFCA	42.9	55.0±3.5	39.8	50.7	30.7	0.95	0.08	4.0	27.0
FeSEM	46.7	59.8±3.2	46.9	55.4	43.7	0.97	0.65	4.0	11.3
MTCFL	66.9	68.8±2.2	59.0	63.7	41.4	0.98	0.19	33.3	0.5
FL+HC	38.8	53.1±2.8	35.9	48.4	34.5	0.94	0.01	4.0	85.1
PACFL	35.3	52.0±2.7	37.2	46.1	29.9	0.95	0.00	4.0	45.4
DisCo	52.5	64.1±2.4	52.3	61.5	48.6	0.98	0.87	4.3	3.0
DisCo-T30	54.7	64.6±2.2	54.9	62.4	49.8	0.99	0.92	5.0	0.1
Oracle	53.0	64.7±2.4	55.5	62.7	49.6	0.99	1.00	4.0	0.0

Table 7: Robustness sweeps (Fashion-MNIST, 3 seeds). For DisCo-CFL: ARI, number of clusters, conflict-merge rate, accuracy and concept accuracy. For comparison: the best baseline on accuracy and on concept accuracy (in parentheses), and the Oracle’s accuracy.

Factor	Setting	DisCo-CFL					Best baseline Acc	Best baseline Concept	Oracle Acc
		ARI	K	Conflict%	Acc	Concept			
Strength	$\theta = 15^\circ$	0.42	3.0	32.1	90.0	82.8	90.2 (FeSEM)	83.0 (FeSEM)	90.5
Strength	$\theta = 30^\circ$	0.93	4.3	1.7	90.3	83.1	90.0 (PACFL)	81.2 (PACFL)	90.4
Strength	$\theta = 45^\circ$	1.00	4.0	0.0	90.4	83.8	89.9 (FeSEM)	81.8 (FeSEM)	90.4
Strength	$\theta = 90^\circ$	1.00	4.0	0.0	90.5	83.9	90.2 (MTCFL)	80.6 (FeSEM)	90.5
Groups	$K = 2$	0.94	3.3	0.0	91.4	86.5	88.5 (IFCA)	80.2 (IFCA)	91.5
Groups	$K = 4$	1.00	4.0	0.0	90.8	84.9	90.6 (FedAvg-FT)	80.9 (FedAvg-FT)	90.8
Groups	$K = 8$	0.92	8.0	0.3	90.4	81.7	88.0 (FeSEM)	75.4 (FeSEM)	90.5
Participation 30%	rotation	1.00	4.0	0.0	89.1	78.4	87.5 (PACFL)	75.1 (FeSEM)	89.1
Participation 30%	label swap	0.98	4.0	0.0	89.3	78.3	86.5 (FeSEM)	73.0 (FeSEM)	89.3
Scale	$N = 200$	0.98	6.3	0.0	91.1	86.6	89.8 (PACFL)	83.9 (FeSEM)	91.1

$\theta \geq 30^\circ$, DisCo-CFL has the best accuracy and concept accuracy and is within 0.1 points of the Oracle.

Number of groups. With $K = 2, 4$ and 8 label-permutation groups, DisCo-CFL finds K without being told (3.3, 4.0 and 8.0 clusters; the extra cluster at $K = 2$ is a conflict-free split) and merges at most 0.3% of conflicting pairs. Its accuracy is within 0.1 points of the Oracle for all K . The margin over the best baseline is 2.9, 0.2 and 2.4 points in accuracy and 6.3, 4.0 and 6.3 points in concept accuracy for $K = 2, 4, 8$. With $K = 2$ and $K = 8$ the baselines given K confuse groups (IFCA and FeSEM merge 25% and 4% of the conflicting pairs), whereas with $K = 4$ the fine-tuned global model is a strong competitor on local accuracy only.

Partial participation. When only 30% of the clients take part in each round, the warm-up model is noisier. DisCo-CFL still recovers the groups (ARI 1.00 and 0.98) and matches the Oracle, while the best baseline is 1.6–2.8 points below in accuracy and 3.3–5.3 points below in concept accuracy.

Number of clients. With 200 clients of 200 samples each, DisCo-CFL reaches ARI 0.98 with no conflicting merge. The extra clusters ($K = 5$ –7) are conflict-free splits of groups; three of them are singletons flagged as anomalous in the audit log. Accuracy equals the Oracle’s (91.1%), 1.3 points above the best baseline, and concept accuracy is 2.7 points above the best baseline. This is in line with the clustering-only scalability study of Section 7.12.

7.5 Statistical significance

Table 8 compares DisCo-CFL with each baseline over all (scenario, seed) pairs with concept shift that both methods were run on. DisCo-CFL is significantly better than every baseline on mean accuracy, concept accuracy and worst-group accuracy (one-sided Wilcoxon signed-rank tests, all $p < 0.01$). The closest competitors are FeSEM, MTCFL and FedAvg-FT. MTCFL and FedAvg-FT win on local accuracy mainly on CIFAR-10 (5–6 of the 6 CIFAR-10 pairs), and FeSEM wins on 8 of 57 pairs, mostly CIFAR-10 rotation and the 15° rotation. On concept accuracy DisCo-CFL wins against every baseline in at least 84% of the pairs.

Table 8: Paired one-sided Wilcoxon signed-rank tests of DisCo-CFL against each baseline over all (scenario, seed) pairs with concept shift. W/T/L: wins, ties ($|\Delta| < 0.05$ points) and losses of DisCo-CFL.

Baseline	pairs	Accuracy		Concept accuracy		Worst-group accuracy	
		W/T/L	p	W/T/L	p	W/T/L	p
FedAvg	57	57/0/0	2.6e-11	57/0/0	2.6e-11	57/0/0	2.6e-11
Local	33	33/0/0	1.2e-10	33/0/0	1.2e-10	32/0/1	2.3e-10
FedAvg-FT	33	27/1/5	2.3e-5	33/0/0	1.2e-10	24/0/9	0.004
IFCA	54	54/0/0	8.1e-11	54/0/0	8.1e-11	54/0/0	8.1e-11
FeSEM	57	45/4/8	2.5e-8	48/2/7	1.4e-8	47/3/7	5.2e-8
MTCFL	39	30/2/7	0.002	39/0/0	1.8e-12	30/0/9	1.4e-4
FL+HC	57	57/0/0	2.6e-11	57/0/0	2.6e-11	57/0/0	2.6e-11
PACFL	57	54/0/3	7.0e-11	57/0/0	2.6e-11	53/1/3	5.1e-11

7.6 Transparency: explanations and diagnosis

Table 9 evaluates the class-level explanations against the ground truth. For label-swap groups, the divergent classes reported by the server are *exactly* the swapped classes in every pair (precision, recall and exact match 1.00), and the diagnosis “class-specific concept shift” is always correct. A typical explanation reads: “*cluster 0 vs. cluster 2: class-specific concept shift on classes [4, 5, 6, 7]*”. Here group 2 swaps $4 \leftrightarrow 5$ and $6 \leftrightarrow 7$. For rotations, precision is 1.00, i.e. no class is ever wrongly reported as divergent. Recall is 0.93: every miss occurs between 0° and 180° groups, on trouser, pullover, dress and shirt, items that are nearly invariant under a 180° rotation. For these classes the two concepts *are* statistically close, so the reported “class-specific shift” matches the data (Proposition 3); it only disagrees with our coarse ground-truth label “global”.

Table 9: Transparency and accountability (Fashion-MNIST, 3 seeds, default settings).

	Rotation	Label swap	Mixed + QS
<i>Explanations: divergent classes between clusters</i>			
precision	1.00	1.00	1.00
recall	0.93	1.00	0.92
exact set / diagnosis correct	0.67	1.00	0.39
<i>Certificates</i>			
share of confident certificates	0.95	0.97	0.93
conflict-free among confident	1.000	1.000	0.919
conflict-free among uncertain	1.000	1.000	0.778
<i>Newcomers</i>			
known concept, correctly assigned	1.00	0.96	–
novel concept, flagged as novel	1.00	1.00	–

7.7 Accountability: certificates, audit log, newcomers

Certificates are informative (Table 9). In the hardest scenario (*Mixed + QS*), 91.9% of the clients with a *confident* certificate are in a conflict-free position, against 77.8% of those with an *uncertain* one. In the other scenarios all clients are correctly placed and 95–97% receive confident certificates. The uncertain ones carry the bottleneck class that makes them ambiguous, which is what a client needs in order to contest its assignment. Newcomers that join after clustering, using the frozen reference model, are assigned to the right cluster in 96–100% of cases. Clients with a labelling never seen during clustering are reported as a *novel concept* in 100% of cases instead of being forced into an existing cluster.

7.8 Privacy

Figure 1 shows the ARI of the clustering when the signatures are released with the Gaussian mechanism of Proposition 2 ($C_{\text{clip}} = 0.5$, $\delta = 10^{-5}$). As predicted by Proposition 2(iii), the *raw* cosine collapses to $\text{ARI} \approx 0$ at every noise level: DP noise attenuates all similarities and every client looks unrelated to every other. With disattenuation, the clustering degrades gracefully, and more data per client buys more privacy, as predicted by the $h \propto \sigma_{\text{dp}}$ term. With 1,500 samples per client, DisCo-CFL keeps ARI 0.92 (rotation) and 0.77 (label swap) at $\varepsilon = 4.4$, and 0.68/0.60 at $\varepsilon = 2$.

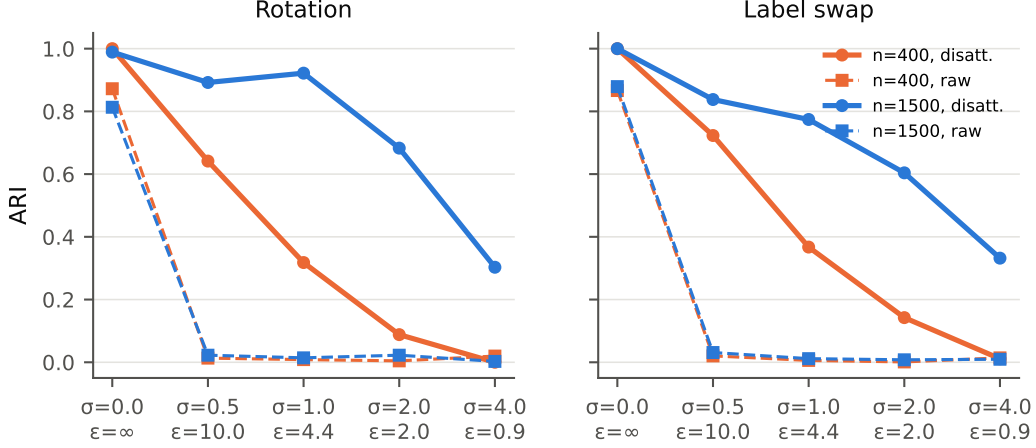


Figure 1: Clustering quality under $(\epsilon, 10^{-5})$ -DP signatures. Solid: DisCo-CFL; dashed: raw cosine. Orange: 400 samples per client; blue: 1,500.

7.9 Calibration and sensitivity

Calibration across sample sizes. Varying the client size from 50 to 800 samples (Figure 4 in Appendix C), the mean disattenuated similarity between clients of the same group stays at 0.98–0.99, whereas the raw cosine stays around 0.91–0.95 and depends on the per-class sample counts. ARI reaches 1.00 from 200 samples per client on (rotation) and from 400 (label swap).

Threshold and shrinkage. Figure 2 shows that DisCo-CFL is stable for $\tau \in [0.75, 0.85]$ in all concept-shift scenarios (ARI 0.85–1.00). With the raw cosine, every τ above 0.7 causes over-splitting (K up to 16 at $\tau = 0.9$). The label-only scenario is the most sensitive: $\tau \leq 0.7$ gives $K = 1$ in all seeds, and at $\tau = 0.8$ one seed flags a single anomalous client. The shrinkage λ has almost no effect between 0.02 and 1 (Appendix C), including values that satisfy the condition of Corollary 1.

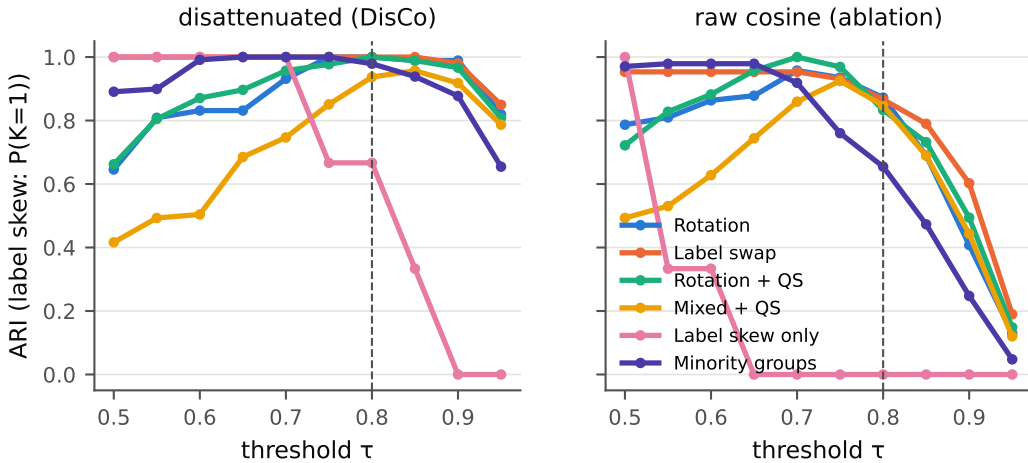


Figure 2: Sensitivity to the threshold τ (3 seeds). The dashed line marks the default $\tau = 0.8$, which was fixed before these runs.

7.10 Theory in practice

Figure 3 validates the theory on synthetic signatures with known ground truth ($p = 512$, noise of effective rank 16). The disattenuated similarity is nearly unbiased for every half size h , whereas the raw cosine is biased by up to -0.37 . Safety and maximality are reached with probability 1 once $h \geq 64$ for all separations $\Delta \in [0.2, 0.7]$, and exact recovery is reached under full class support. The required h grows as Δ shrinks. Recovery of a minority group of 1, 2, 4 or 8 clients follows the same curve (Figure 5 in Appendix C), as Corollary 1 states.

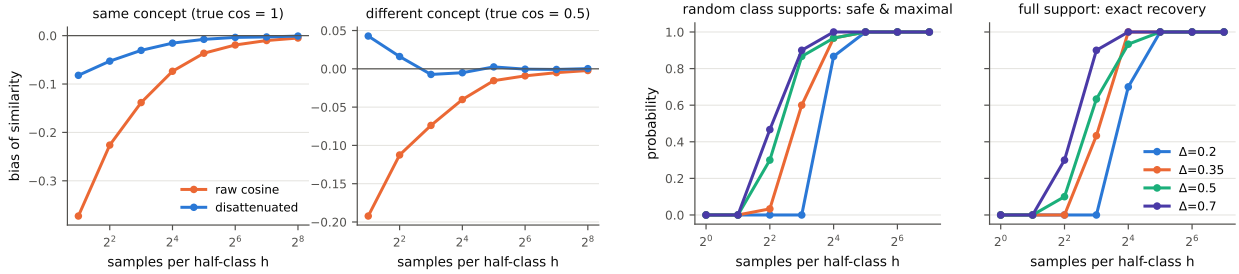


Figure 3: Synthetic validation. Left: bias of the similarity estimate. Right: probability of the guarantees of Theorem 2.

7.11 Ablation

Table 10 removes one component at a time. Every component contributes. Without class conditioning (one signature per client over all its data), clients are grouped by their label mix, exactly the failure mode reported in the survey [Ben Ali et al., 2026]: 13–23 clusters and a concept accuracy 10–12 points lower. Without disattenuation, noise-attenuated similarities fall below the threshold and the federation is over-split, including the label-only case. The mean linkage averages class-specific shifts away and merges conflicting clients (up to 6.9% in *Mixed + QS*). Linkage-ordered merging creates bridges through ambiguous clients and over-splits. Without clipping, heavy-tailed per-sample gradients inflate the number of clusters.

Table 10: Ablation (Fashion-MNIST, 3 seeds): ARI / number of clusters K per scenario, mean conflict-merge rate and mean concept accuracy over the five scenarios.

Variant	Rotation	Label swap	Rot. + QS	Mixed + QS	Label only	Conflict%	Concept
DisCo-CFL (full)	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	0.94 / 4.0	0.67 / 1.3	0.1	83.8
without disattenuation (raw cosine)	0.87 / 6.7	0.87 / 6.0	0.83 / 7.0	0.85 / 5.7	0.00 / 4.0	0.0	82.4
without class conditioning	0.15 / 23.3	0.15 / 17.7	0.16 / 22.3	0.09 / 22.3	0.00 / 13.0	0.9	72.7
mean instead of bottleneck linkage	1.00 / 4.0	0.80 / 4.0	0.92 / 4.0	0.69 / 4.0	1.00 / 1.0	2.6	82.1
linkage-ordered merging	0.95 / 5.0	0.86 / 5.7	1.00 / 4.0	0.81 / 5.7	0.33 / 1.7	0.0	82.5
without per-sample clipping	0.94 / 5.0	1.00 / 4.0	0.91 / 5.7	0.88 / 5.3	0.00 / 3.0	0.3	83.0

7.12 Cost and scalability

Each client uploads its signature *once*, about 1.2×10^4 floats with $p = 1024$. The size is independent of the model dimension d ; IFCA, by contrast, downloads K models per client in *every* round, and CFL/FL+HC cluster d -dimensional updates. Computing the signature takes 0.14–0.16s per client on one CPU core. Server-side clustering takes 0.05s for 40 clients and 2.3s for 320 clients, with $\text{ARI} \geq 0.98$ at every scale.

8 Discussion and limitations

What is new. Earlier CFL methods cluster whatever the clients send (updates, losses, data statistics) with a similarity whose scale depends on sample size, on training progress and on noise. The number of clusters then depends on a hyperparameter that has no fixed meaning. DisCo-CFL defines the clustering target precisely (class-conditional concept equivalence) and estimates a similarity whose population value is known in advance for clients that share a concept, namely 1. The invariance, the automatic choice of K , the size-independent recovery, the robustness to DP noise and the explanation guarantees are all consequences of this calibration.

Limitations. (i) The class support S_i of each client and the per-class half sizes are revealed to the server; the DP guarantee protects individual records given the labels, not the label histogram. (ii) The analysis assumes independent sub-Gaussian per-sample signatures and a fixed reference model. It does not cover the attachment heuristics, and the constants are not optimised. The upper bound exceeds the lower bound by a factor $1/\Delta$. (iii) Concept shifts that leave the reference-model gradient unchanged are invisible by construction (Assumption 2). An example is a 180° rotation of nearly symmetric Fashion-MNIST items; for classification such shifts are also largely harmless. (iv) The signatures are only as informative as the reference model. On CIFAR-10 a 10-round warm-up gives a weak model and weak separation, and 30 rounds are needed for good clustering. An adaptive warm-up that stops when the split-half reliabilities and the between-cluster gaps stabilise is a natural extension. (v) Clients that hold none of the discriminative classes are compatible with several groups. DisCo-CFL may place them in any of these groups, which is harmless for training but lowers ARI. (vi) Clustering is one-shot at the reference model. Concept drift after clustering would require recomputing signatures, although the newcomer procedure already detects clients whose concept is unseen. (vii) Our experiments use small CNNs and synthetic concept groups built from public image datasets, as is standard in the CFL literature [Ben Ali et al., 2026]. Natural federated datasets and larger models remain to be tested.

9 Conclusion

DisCo-CFL replaces ad hoc similarity heuristics in clustered FL with a *calibrated* statistic: the disattenuated cosine between class-conditional gradients. The statistic is invariant to label and quantity skew and stays consistent under privacy noise. Its threshold has a fixed meaning, which gives an automatic K , and the resulting clustering has recovery guarantees that do not depend on group size and explanations whose fidelity is proven. The algorithm reports assignment certificates, an audit log, anomaly flags and class-level explanations together with the clusters, which extends the evaluation of CFL from accuracy alone to fairness, accountability, transparency and privacy.

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A Proofs

Throughout, $C, C_1, C_2, \dots$ and c_0 denote constants that depend only on the sub-Gaussian constant K_0 and may change from line to line. We fix a class c and drop it from the notation when no confusion arises. Statements whose deterministic core is machine-checked in Lean 4 are marked with the name of the corresponding Lean theorem (Appendix B).

A.1 Proof of Proposition 1

The reference model w , the sketch Φ and the label c are the same for all clients, so

$$\mu_i^c = \int \Phi \nabla_w \ell(w; x, c) d\mathcal{P}_i(x \mid Y = c)$$

is determined by $\mathcal{P}_i(X \mid Y = c)$ alone. Neither the proportion $\mathcal{P}_i(Y = c)$ nor the number of samples enters. If i and j are compatible, then $\mathcal{P}_i(X \mid Y = c) = \mathcal{P}_j(X \mid Y = c)$ on each shared class, hence $\mu_i^c = \mu_j^c$ and $\rho_{ij}^c = 1$. Per-sample clipping is applied identically by all clients, so the same holds for the expected clipped signature. $\square$

A.2 Proof of Lemma 1

Write the half-means as $a_i = \mu_i + \zeta_{i1}$ and $b_i = \mu_i + \zeta_{i2}$, where ζ_{i1}, ζ_{i2} are independent, zero-mean and have covariance Σ_i/h_i (any covariance works, and it may differ across clients). Let $t_i = \text{tr } \Sigma_i/h_i$. Then

$$\mathbb{E}\langle a_i, b_i \rangle = a_i, \quad \mathbb{E}\|a_i\|^2 = \mathbb{E}\|b_i\|^2 = a_i + t_i,$$

so the population split-half reliability is $r_i^* = a_i/(a_i + t_i)$ and

$$R_i^* = \frac{2r_i^*}{1 + r_i^*} = \frac{a_i}{a_i + t_i/2}.$$

For the full means, $\mathbb{E}\|m_i\|^2 = a_i + t_i/2 = a_i/R_i^*$, and by independence across clients $\mathbb{E}\langle m_i, m_j \rangle = \langle \mu_i, \mu_j \rangle$. The population cosine is therefore

$$\frac{\langle \mu_i, \mu_j \rangle}{\sqrt{(a_i/R_i^*)(a_j/R_j^*)}} = \frac{\langle \mu_i, \mu_j \rangle}{\sqrt{a_i a_j}} \sqrt{R_i^* R_j^*} = \rho_{ij} \sqrt{R_i^* R_j^*},$$

and dividing by $\sqrt{R_i^* R_j^*}$ gives ρ_{ij} exactly. (Lean: `spearman_brown, population_calibration`.) $\square$

A.3 Proof of Theorem 1

Noise representation. A half-mean noise vector is $\zeta = (\Sigma/h)^{1/2}\bar{z}$ with $\bar{z} = h^{-1/2}\sum_s z_s$. Its coordinates are independent, zero-mean, unit-variance and CK_0 -sub-Gaussian.

Statistics. $\hat{\rho}_{ij}$ and R_i are functions of nine statistics:

$$X_1 = \langle a_i, b_i \rangle, \quad X_2 = \|a_i\|^2, \quad X_3 = \|b_i\|^2, \quad X_4 = \|m_i\|^2,$$

the same four statistics $X_5, \dots, X_8$ for client j , and $X_9 = \langle m_i, m_j \rangle$. We normalise them as $x_k = X_k/a_i$ for the statistics of i , $x_k = X_k/a_j$ for those of j , and $x_9 = X_9/\sqrt{a_i a_j}$. With $\kappa_i = \text{tr } \Sigma/(a_i h_i) \leq \kappa$, their expectations are

$$x^* = (1, 1 + \kappa_i, 1 + \kappa_i, 1 + \kappa_i/2, \dots, \rho_{ij}).$$

Each statistic is a constant, plus linear terms $\langle \mu, \zeta \rangle$, plus quadratic or bilinear forms in the ζ 's.

Linear terms. $\langle \mu_i, \zeta \rangle$ is sub-Gaussian with variance proxy

$$CK_0^2 \frac{\mu_i^\top \Sigma \mu_i}{h} \leq CK_0^2 \frac{a_i \|\Sigma\|_{\text{op}}}{h}.$$

After normalisation by a_i (or by $\sqrt{a_i a_j}$ for cross terms), it is, with probability $1 - \delta'$, at most

$$C \sqrt{\frac{\|\Sigma\|_{\text{op}}}{ah}} \sqrt{\log(2/\delta')} = C \sqrt{\kappa/r_s} \sqrt{\log(2/\delta')},$$

because $\|\Sigma\|_{\text{op}} = \sigma^2/r_s$.

Quadratic and bilinear terms. $\|\zeta\|^2 - \text{tr}(\Sigma/h)$ and $\langle \zeta, \zeta' \rangle$ for independent ζ, ζ' are centred quadratic forms $\bar{Z}^\top M \bar{Z}$ of the concatenated vector $\bar{Z}$, with $\|M\|_F \leq \|\Sigma\|_F/h$ and $\|M\|_{\text{op}} \leq \|\Sigma\|_{\text{op}}/h$. The Hanson–Wright inequality [Rudelson and Vershynin, 2013] bounds the deviation, with probability $1 - \delta'$, by

$$CK_0^2 \left(\|M\|_F \sqrt{\log(2/\delta')} + \|M\|_{\text{op}} \log(2/\delta') \right).$$

Since $\|\Sigma\|_F = \sigma^2/\sqrt{r_e}$, after normalisation by a this is at most

$$C \frac{\kappa}{\sqrt{r_e}} \sqrt{\log(2/\delta')} + C \frac{\kappa}{r_s} \log(2/\delta'),$$

and the second term is at most $C\eta^2 \leq Cc_0\eta$.

Union bound. Taking $\delta' = \delta/6$ over the at most six independent noise terms that appear in each group of statistics, we obtain $\|x - x^*\|_\infty \leq C\eta$ with probability $1 - \delta$.

Smoothness. Write $\hat{\rho} = F(x)$, where, before clipping,

$$r_i = \frac{x_1}{\sqrt{x_2 x_3}}, \quad R_i = \frac{2r_i}{1 + r_i} \quad (\text{similarly for } j), \quad \hat{\rho} = \frac{x_9}{\sqrt{x_4 x_8 R_i R_j}}.$$

By Lemma 1, $F(x^*) = \rho_{ij}$. For $\kappa \leq 1$ and $\|x - x^*\|_\infty \leq Cc_0$ with c_0 small enough, the argument stays in a compact set on which $x_2, x_3, x_4 \in [1/2, 3]$, $r_i \geq 1/5$ and $R_i \geq 1/3$. On that set F is continuously differentiable with gradient bounded by an absolute constant. Hence $|F(x) - F(x^*)| \leq C_1\eta$, and the same argument gives the bound for R_i . Clipping to $[-1, 1]$ can only reduce the error, because $\rho_{ij} \in [-1, 1]$ (Lean: `clip_reduces_error`). For the raw cosine,

$$\frac{x_9}{\sqrt{x_4 x_8}} \rightarrow \frac{\rho_{ij}}{\sqrt{(1/R_i^*)(1/R_j^*)}} = \rho_{ij} \sqrt{R_i^* R_j^*}. \quad \square$$

A.4 Proof of Theorem 2

We use two facts, both on $\mathcal{E}$.

Claim 1 (conflicting clusters cannot be linked). Let A, B be conflict-free and suppose some $i \in A$ and $j \in B$ conflict on class c . Since A is conflict-free, all members of A holding c share one class-conditional $\mathcal{P}_A^c$; likewise for B , and $\mathcal{P}_A^c \neq \mathcal{P}_B^c$. Hence *every* pair in $A \times B$ holding c conflicts on c . By Assumption 2 and $\mathcal{E}$, each such pair has $\hat{\rho} \leq 1 - \Delta + \gamma/2$, so

$$\text{num}_{AB}^c \leq (1 - \Delta + \gamma/2) \text{den}_{AB}^c.$$

Moreover

$$\frac{\text{num}_{AB}^c + \lambda}{\text{den}_{AB}^c + \lambda} < \tau \iff \text{den}_{AB}^c (\tau - 1 + \Delta - \gamma/2) > \lambda(1 - \tau).$$

Since $\tau - 1 + \Delta \geq \gamma$, the left side is at least

$$\text{den}_{AB}^c \frac{\gamma}{2} \geq W_{ij}^c \frac{\gamma}{2} > w_0 \frac{\gamma}{2} = \lambda(1 - \tau).$$

The minimum over classes in (3) is therefore below τ . (Lean: `claim1_conflict_not_linked`.)

Claim 2 (compatible clusters are always linkable). If $A \cup B$ is conflict-free, then every pair in $A \times B$ that holds a class c has $\rho^c = 1$ (Proposition 1), so $\hat{\rho}^c \geq 1 - \gamma/2$. Every class term of (3) is then at least

$$1 - \frac{\gamma}{2} \geq 1 - \frac{1 - \tau}{2} = \frac{1 + \tau}{2} > \tau,$$

and classes without evidence contribute 1. So $L(A, B) > \tau$ whenever the shared evidence is at least $e_{\min}$. (Lean: `claim2_compatible_linked`.)

Conclusion. (a) Singletons are conflict-free. By Claim 1, a merge is only performed between clusters whose union is conflict-free, so the invariant is preserved by every merge, in any order. (b) At termination, every pair of clusters with shared evidence $\geq e_{\min}$ has $L < \tau$; by Claim 2 their union is not conflict-free. (c) Under full support, two clients of different groups $k \neq l$ both hold a class on which $\mathcal{P}^{(k)}$ and $\mathcal{P}^{(l)}$ differ, so they conflict. By (a), each output cluster is contained in a single group. Two output clusters inside the same group have a conflict-free union and share evidence $\geq e_{\min}$, which contradicts (b). Each group thus forms exactly one cluster. (Lean: `merge_preserves_conflict_free, safety_all_merges, exact_recovery`.) $\square$

A.5 Proof of Corollary 1

There are at most $N^2 C/2$ pairwise class estimates and NC reliabilities; we apply Theorem 1 to each with $\delta/(N^2 C)$. With $\mathsf{L} = \log(12N^2 C/\delta)$, the requirement $C_1 \eta \leq \gamma/2$ holds if

$$\sqrt{\kappa/r_s} \sqrt{\mathsf{L}} \leq \frac{\gamma}{4C_1} \quad \text{and} \quad \frac{\kappa}{\sqrt{r_e}} \sqrt{\mathsf{L}} \leq \frac{\gamma}{4C_1}.$$

With $\kappa = \sigma^2/(ah)$ these are the first two terms of (5) (Lean: `sample_size_sufficient`). The side conditions $\kappa \leq 1$ and $\eta \leq c_0$ are absorbed into C_2 and the third term. For the evidence condition: on the event, $R \geq R^* - \gamma/2 \geq 2/3 - \gamma/2$ because $\kappa \leq 1$. For $\gamma \leq 1/3$ this gives $W \geq 1/4$, so $W > w_0$ holds whenever $\lambda < \gamma/(8(1 - \tau))$ (Lean: `evidence_condition`). None of these conditions involves $|G_k|$. $\square$

A.6 Proof of Theorem 3

Let e_1 be a top eigenvector of Σ ($\Sigma e_1 = \|\Sigma\|_{\text{op}} e_1$) and let $e_2 \perp e_1$ be a unit vector. With $s^2 = \Delta/2$ put

$$u = \sqrt{a}(\sqrt{1-s^2}e_2 + se_1), \quad v = \sqrt{a}(\sqrt{1-s^2}e_2 - se_1).$$

Then

$$\|u\|^2 = \|v\|^2 = a, \quad \cos(u, v) = 1 - 2s^2 = 1 - \Delta, \quad u - v = 2\sqrt{a}se_1$$

(Lean: `lower_bound_construction`). Under H_0 the n samples of client j are i.i.d. $\mathcal{N}(u, \Sigma)$; under H_1 they are $\mathcal{N}(v, \Sigma)$; $\mu_i = u$ is known. Then

$$\text{KL}(\mathcal{N}(u, \Sigma)^{\otimes n} \parallel \mathcal{N}(v, \Sigma)^{\otimes n}) = \frac{n}{2}(u - v)^\top \Sigma^{-1}(u - v) = \frac{n}{2} \cdot \frac{4as^2}{\|\Sigma\|_{\text{op}}} = \frac{na\Delta r_s}{\sigma^2}.$$

By the Bretagnolle–Huber inequality (Tsybakov, 2009, Thm. 2.2), every test ψ satisfies

$$\max_{H \in \{0,1\}} \Pr(\psi \neq H) \geq \frac{1}{4} \exp(-\text{KL}).$$

A procedure that sees only half-means (as DisCo-CFL does), or that does not know μ_i , has access to less information, so the bound applies to it as well. $\square$

A.7 Proof of Proposition 4

For i in cluster k , since the class-conditionals are shared,

$$\mathcal{P}_i(y | x) = \frac{\mathcal{P}_i(y) \mathcal{P}(x | y)}{\sum_{y'} \mathcal{P}_i(y') \mathcal{P}(x | y')}, \quad q_k(y | x) = \frac{\mathcal{P}_k(y) \mathcal{P}(x | y)}{\sum_{y'} \mathcal{P}_k(y') \mathcal{P}(x | y')}.$$

Dividing, $\mathcal{P}_i(y | x) \propto q_k(y | x) \mathcal{P}_i(y) / \mathcal{P}_k(y)$, with a normalising constant that does not depend on y . Taking logarithms and the argmax gives (4) (Lean: `prior_correction_proportional`, `prior_correction_argmax`). The Bayes classifier of i is thus a function of q_k and $\mathcal{P}_i(Y)$ only. Splitting the cluster changes neither quantity, so it cannot reduce the Bayes risk; it only reduces the data available to estimate q_k . $\square$

A.8 Proof of Proposition 2

(i) Replacing one record by another of the same label changes one clipped summand. The half-mean therefore changes by at most $2C_{\text{clip}}/h$ in ℓ_2 (Lean: `clipped_mean_sensitivity`). The added noise has standard deviation σ_{dp} times this sensitivity, so the analytic Gaussian mechanism [Balle and Wang, 2018] gives (ε, δ) -DP for that vector. Every record enters exactly one released vector, so by parallel composition the whole signature is (ε, δ) -DP. The server’s computations are post-processing [Dwork and Roth, 2014]. The class supports and half sizes are released and are not protected.

(ii) The DP noise is zero-mean and independent of everything else. The proof of Lemma 1 used only these two properties, so it still holds with

$$t_i = \frac{\text{tr } \Sigma_{\text{clip}}}{h} + p \left(\frac{2\sigma_{\text{dp}} C_{\text{clip}}}{h} \right)^2.$$

In the proof of Theorem 1, the Gaussian DP component adds a linear term with normalised standard deviation $2\sigma_{\text{dp}} C_{\text{clip}} / (h\sqrt{a})$ and a quadratic term with normalised Frobenius scale $4\sqrt{p} \sigma_{\text{dp}}^2 C_{\text{clip}}^2 / (h^2 a)$. Requiring each to be at most $\gamma / (4C_1 \sqrt{L})$ gives the two stated conditions.

(iii) This follows from the last statement of Theorem 1 with the inflated t_i . $\square$

A.9 Proof of Proposition 3

Output clusters are conflict-free (Theorem 2(a)), so the members of A holding c share $\mathcal{P}_A^c$, and likewise for B . If $\mathcal{P}_A^c \neq \mathcal{P}_B^c$, the computation of Claim 1 with $\text{den}_{AB}^c > w_0$ gives a pooled value below τ , so $c \in \hat{D}_{AB}$. Otherwise all pairs have $\rho^c = 1$ and the pooled value is at least $(1 + \tau)/2 > \tau$ (Claim 2), so $c \notin \hat{D}_{AB}$. (Lean: `explanation_fidelity`.) $\square$

B Lean 4 verification

We formalised the deterministic core of every result of Section 5 in Lean 4 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The project is in the `lean/` directory of the repository https://github.com/vinhqdang/cluster_federated_learning and is checked with `lake build`. None of the 25 theorems uses `sorry`, and `#print axioms` reports only Lean’s standard axioms (`propext`, `Classical.choice`, `Quot.sound`) for each of them. Table 11 maps the results to Lean theorems.

The probabilistic ingredients enter the formal statements as hypotheses: sub-Gaussian and Hanson–Wright concentration, the Bretagnolle–Huber inequality and the (ε, δ) accounting of the Gaussian mechanism. For example, `claim1_conflict_not_linked` assumes the event $\mathcal{E}$ in the form $\text{num} \leq (1 - \Delta + \gamma/2) \text{den}$. These ingredients are standard, cited results that Mathlib does not yet provide in the required form, so they are not re-proved. The combinatorial part of Theorem 2 is modelled abstractly. A state is a list of clusters and a merge step replaces two linked clusters by their union. Safety is proved for every state reachable by *any* sequence of allowed merges (`Relation.ReflTransGen`), so it does not depend on the merge order.

Table 11: Results of Section 5 and the Lean theorems that verify their deterministic core.

Result	Lean theorems (namespace <code>DiscoProofs</code>)
Lemma 1	<code>spearman_brown</code> , <code>population_calibration</code> , <code>raw_cosine_attenuated</code> , <code>attenuation_le_one</code>
Theorem 1 (clipping step)	<code>clip_reduces_error</code>
Theorem 2, Claims 1–2	<code>claim1_conflict_not_linked</code> , <code>claim2_compatible_linked</code> , <code>claim2_lower_bound</code>
Theorem 2(a) safety	<code>singleton_conflict_free</code> , <code>merge_preserves_conflict_free</code> , <code>safety_all_merges</code>
Theorem 2(b) maximality	<code>maximality</code>
Theorem 2(c) exact recovery	<code>exact_recovery</code>
Corollary 1	<code>sample_size_sufficient</code> , <code>evidence_condition</code>
Theorem 3	<code>lower_bound_construction</code> , <code>lower_bound_cosine</code> , <code>kl_value</code> , <code>lower_bound_sample_size</code>
Proposition 2(i)	<code>norm_clip_le</code> , <code>clipped_mean_sensitivity</code> , <code>clipped_signature_sensitivity</code>
Proposition 3	<code>explanation_fidelity</code>
Proposition 4	<code>prior_correction_proportional</code> , <code>prior_correction_argmax</code>

C Additional results

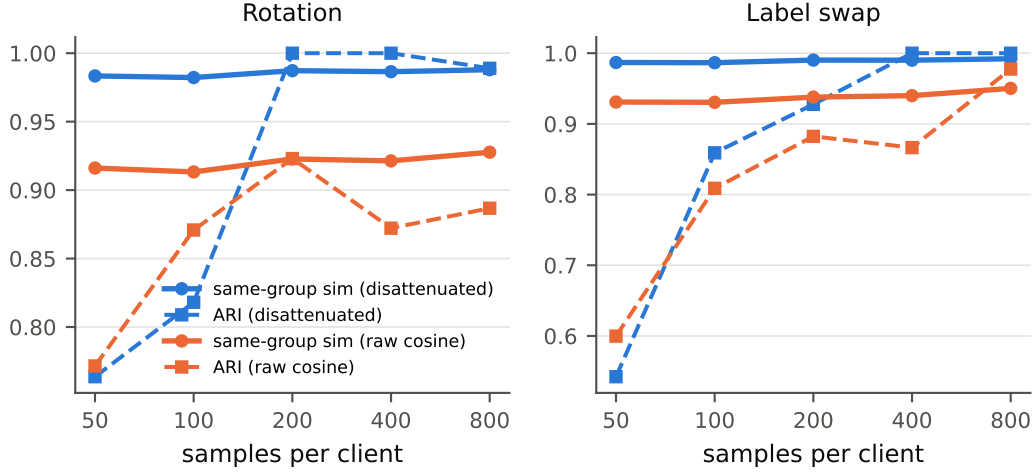


Figure 4: Same-group similarity (solid) and ARI (dashed) as a function of the client size. Blue: disattenuated; orange: raw cosine.

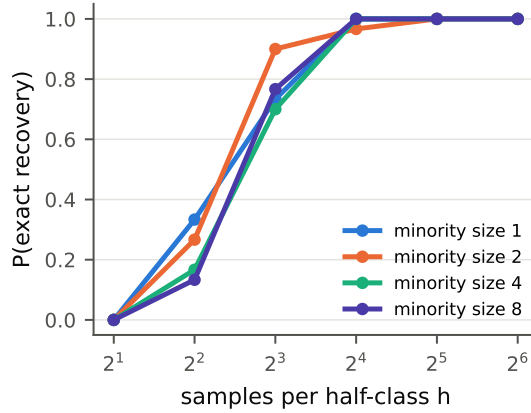


Figure 5: Synthetic validation of Corollary 1: exact recovery of a minority group does not depend on its size.

Table 12: Sensitivity to the shrinkage λ at $\tau = 0.8$: ARI / number of clusters (3 seeds, clustering step). For the label-only scenario the correct K is 1.

Scenario	$\lambda = 0.02$	$\lambda = 0.1$	$\lambda = 0.2$	$\lambda = 0.5$	$\lambda = 1.0$
Rotation	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	0.98 / 4.0
Label swap	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0	1.00 / 4.0
Rotation + QS	0.99 / 4.3	0.99 / 4.3	0.99 / 4.3	1.00 / 4.0	0.98 / 4.0
Mixed + QS	0.96 / 4.0	0.94 / 4.0	0.94 / 4.0	0.94 / 4.0	0.91 / 4.0
Label skew only	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3	0.67 / 1.3
Minority groups	0.98 / 4.3	0.98 / 4.3	0.98 / 4.3	0.98 / 4.3	1.00 / 4.0